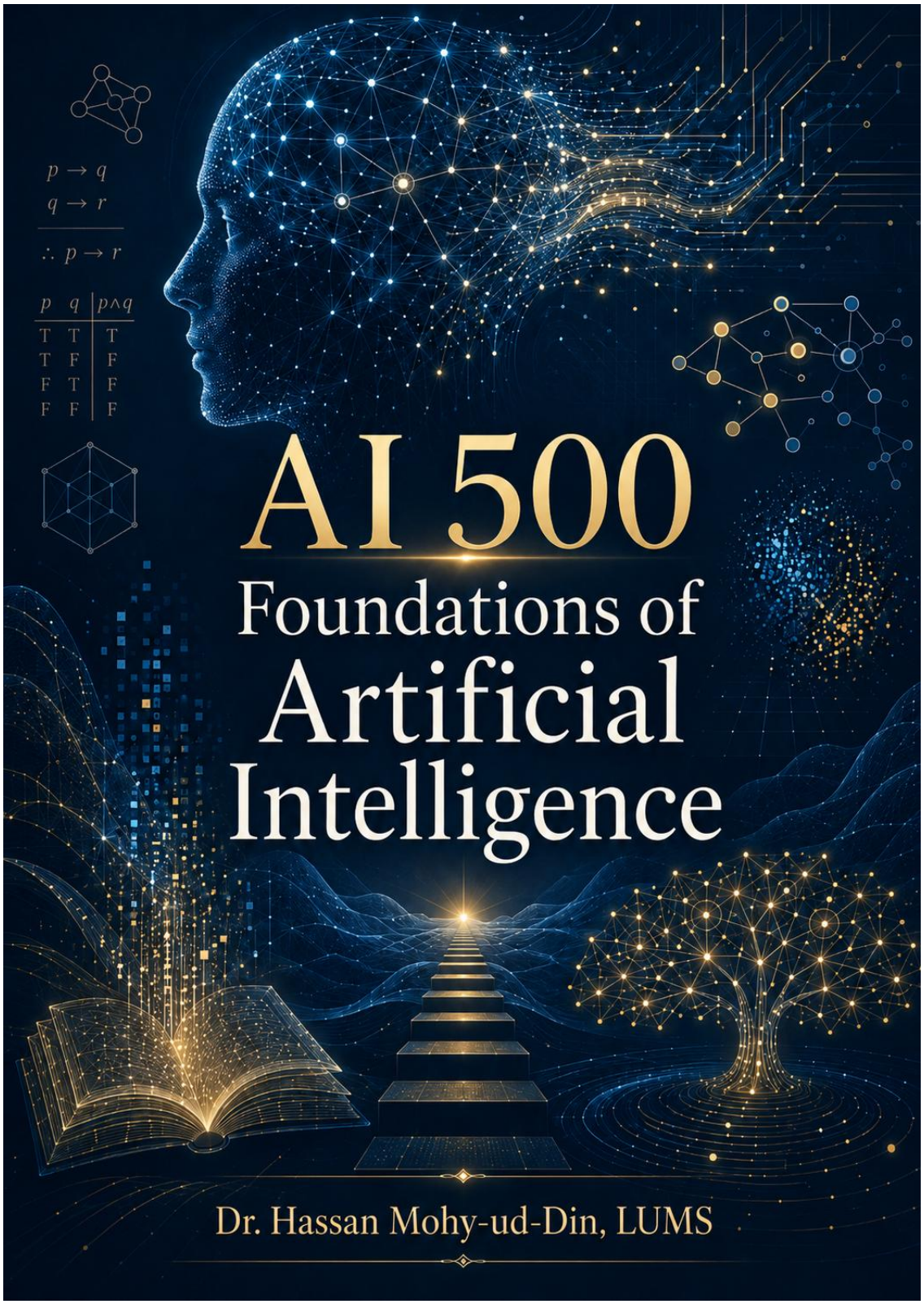



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# About the Cover Image

The cover image presents artificial intelligence as a journey from foundational knowledge to purposeful innovation. The luminous human profile, interwoven with neural networks, mathematical symbols, logical structures, and computational pathways, represents the union of human reasoning and machine intelligence. The open book symbolizes scholarship and the enduring intellectual foundations of the field, while the ascending staircase suggests progressive learning – from data representation, probability, optimization, and algorithms to machine learning, evaluation, deployment, and responsible decision-making. The networked tree signifies the growth of knowledge, the interconnectedness of AI disciplines, and the emergence of increasingly sophisticated systems from fundamental principles. Together, these elements reflect the central philosophy of this course is not merely a collection of algorithms, but a complete, evolving discipline grounded in mathematics, data, computation, judgment, ethics, and real-world application.

## Disclaimer

In the compilation of this document, I have utilized GPT-based tools, including ChatGPT 5.6 Sol, to assist with the generation of cover and module images, schematics, flowcharts, infographics, tables, workflows, and synthetic examples, as well as with the identification and synthesis of published references, historical information, and other relevant background material. These tools were also used to support the organization, refinement, and presentation of the material. All generated content and references were reviewed, edited, and, where appropriate/possible, independently verified by the author. Nevertheless, errors, omissions, or inaccuracies may remain, and I would be grateful to readers for identifying and bringing any such issues to my attention so that they may be corrected in future revisions.

I do not harbor unbridled enthusiasm or fascination of AI. I firmly believe that technology should remain *semi-autonomous* i.e., operating under informed judgment, ethical oversight, and deliberate control of a wise human mind. In this spirit, AI is treated here as a supportive tool rather than a substitute for human reasoning, creativity, or responsibility.

# About the Instructor

Dr. Hassan Mohy-ud-Din is the Director of Algorithms in Theory and Practice Lab and an Assistant Professor of Electrical Engineering, Syed Babar Ali School of Science and Engineering, LUMS. He completed his PhD and MSE in Electrical and Computer Engineering and MA in Applied Mathematics and Statistics from Johns Hopkins University (2009 – 2015). He was recipient of PhD Fellowship from Johns Hopkins University. From 2015 – 2017 he was a postdoctoral associate in the Department of Radiology and Biomedical Imaging at the Yale School of Medicine. From 2017 – 2018 he was a Clinical Research Scientist at Shaukat Khanum Memorial Cancer Hospital and Research Centre, Lahore, Pakistan. His research is at the intersection of applied mathematics and clinical imaging – exploiting tools in machine learning, optimization, statistics, and information theory to develop novel algorithms for clinical and translational imaging. He has done extensive research in multimodality imaging including PET/CT, SPECT/CT, PET/MR, Low-dose CT, and multiparametric MRI and developed computational pipelines for brain imaging, cardiac imaging, and abdominal imaging. His work on dynamic cardiac PET imaging won the 2014 SNMMI Bradley-Alavi fellowship and the 2014 SIAM student award. He is also a recipient of the 2019 Charles Wallace Fellowship from the British Council, Pakistan. He also serves as a reviewer on major scientific journals and a member of IEEE and SIAM societies. He carries a university teaching experience of over fifteen years (including five years at Johns Hopkins University) and is deeply interested in the development, implementation, and evaluation of (pragmatic and effective) pedagogical approaches in schools, colleges, and universities.




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# Data as a Constructed Record of Reality

The phrase **raw data** is a dangerous oxymoron. In machine learning and data science, it is a **common** but **critical mistake** to treat **datasets as objective**, **pre-existing truths** that simply exist in the world, waiting to be **passively harvested**. In reality, data is **never truly raw**; it is always *cooked*. Before a single row is ever logged in a database, it has already been **shaped by human assumptions**, **technical constraints**, and **systemic priorities**. To understand how an AI system will ultimately behave, we must first recognize that data is **not a passive reflection** of the world, but an **active**, **engineered artifact**.

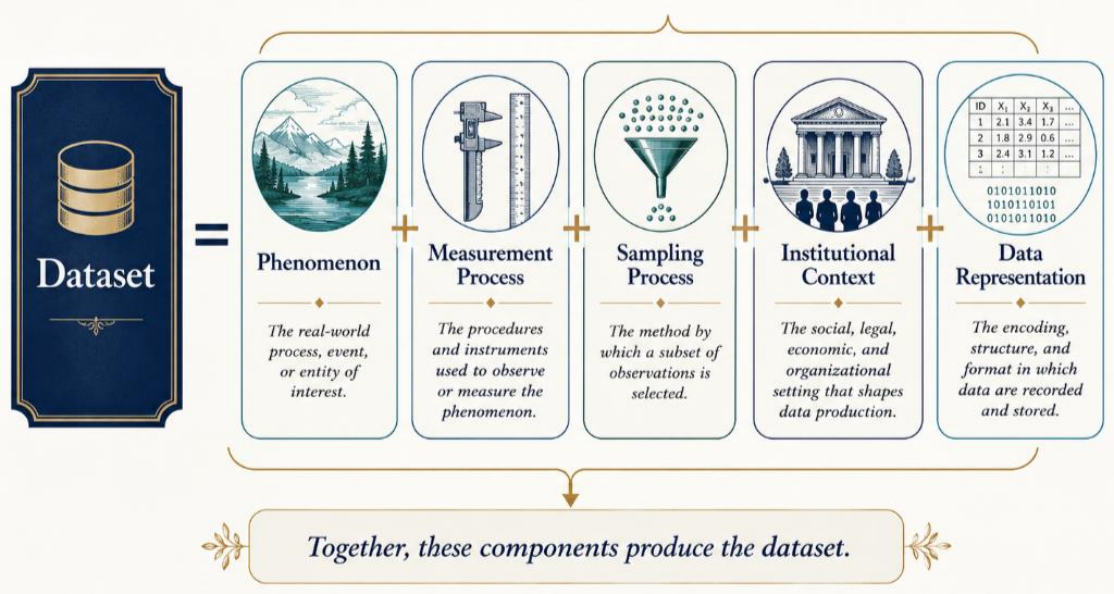



**Figure 14: Conceptual formulation of a dataset.**

## 2.1.1 Data are produced, not simply found

Every dataset is **actively produced** through a **long chain of subjective**, **technical**, and **systemic choices**. Before a single row is entered into a database, decisions must be made about:

- **Conceptualization:** What phenomenon actually matters? How will variables and labels be strictly defined?
- **Observation:** What is the unit of analysis? Who or what will be included or excluded? Which instruments or sensors will be used, and how frequently will measurements be taken?

- **Engineering:** Which observations will actually be recorded? How will missing data be encoded? How will the data be stored, linked, and transformed?

Because of this **complex genesis**, a dataset is better understood through the formulation depicted in **Figure 14**.

The values in a spreadsheet may look like precise, objective integers, but the **underlying concepts** they represent often **remain deeply ambiguous**. A database column labeled *disease*, *fraud*, *dropout*, *risk*, or *success* is **never self-defining**. Its true meaning is **entirely dictated** by the **operational rules**, **human biases**, and **institutional practices** used to generate it.

To combat this opacity, researchers proposed **Datasheets for Datasets**[196] to **force transparency**. A datasheet rigorously documents a dataset's underlying **motivation**, **composition**, **collection mechanics**, **preprocessing steps**, **intended uses**, **distribution**, and **maintenance**, ensuring that future users understand exactly how the **reality was constructed**.

### 2.1.2 Fitness for Purpose

There is **no such thing** as a **universally good** or **high-quality dataset** in a vacuum. A dataset is **only good** for a **specific purpose** under **specific conditions**.

Data quality is **fundamentally contextual**. For example, a dataset might be:

- Adequate for descriptive statistics but **fatally flawed for causal inference**.
- Useful for retrospective research but entirely **unsuitable for real-time prediction**.
- Representative of a specialized urban hospital but **useless for a rural clinic**.
- Sufficient for adult populations but **invalid for pediatric care**.
- Appropriate for binary classification but **poorly calibrated for absolute risk estimation**.

[196] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford, *Datasheets for Datasets*, arXiv, 2018.

A useful conceptual formulation to evaluate this is:

$$Q(D, T, C) = \text{The fitness of Dataset } D \text{ for Task } T \text{ under Context } C$$

This aligns perfectly with NIST guidelines, which **evaluate data quality** and **representativeness** strictly in relation to the **intended use**, emphasizing **practical dimensions** such as **accuracy**, **completeness**, **consistency**, **relevance**, and **timeliness**.

Hence, never ask, **Is this data good?** Always ask, **Is this data good for the specific task I am trying to accomplish right now?** Data is not a flawless mirror reflecting reality; it is a constructed map. If we do not understand how the map was drawn, and whether it is scaled for our specific journey, we will inevitably get lost.

*Example # 38: Predicting student dropout risk*

Suppose a university wants to build an AI system that predicts which students are at risk of dropping out so advisors can intervene early.

*Step 1: Defining the problem*

At first glance, **dropout** seems straightforward. But the project team must decide **what it actually means**. Possible definitions include:

- A student who permanently leaves the university.
- A student who fails to enroll the following semester.
- A student who transfers to another institution.
- A student who takes a temporary leave of absence.
- A student who graduates late.

**Each definition** produces a **different dataset** because different students receive different labels. This illustrates that the **target variable** is **constructed**, not discovered.

*Step 2: Choosing what to measure*

Next, the university decides which information to collect. **Table 4** includes few variables collected for the study. Already, the dataset represents only a **partial view of reality**. Students experiencing severe personal problems may appear

perfectly healthy in the data simply because **those problems were never measured**.

**Table 4: Inclusion and exclusion of candidate variables for student-risk modeling.** The table documents the feature-selection rationale for a student-focused predictive model by identifying which variables were included or excluded and the justification for each choice. Included variables reflect academically, behaviorally, or operationally relevant information, whereas excluded variables were omitted because of privacy constraints or insufficient data reliability.

| Variable | Included? | Why? |
|---|---|---|
| GPA | ✓ | Academic performance |
| Attendance | ✓ | Engagement |
| Library usage | ✓ | Study habits |
| Tuition payment history | ✓ | Financial stress |
| Mental health counseling visits | X | Privacy restrictions |
| Part-time employment | X | No reliable records |
| Family income | ✓ | Available from financial aid forms |

*Step 3: Sampling decisions*

The university collects data only from students enrolled on the main campus. Excluded are:

- Online students
- Exchange students
- Graduate students
- Students from satellite campuses

Now the dataset is **representative only of one population**. Applying the resulting model nationwide would likely produce unreliable predictions.

*Step 4: Recording the data*

Suppose attendance is collected through electronic classroom check-ins.

- But some instructors never use the scanner.
- Others record attendance manually.
- Some classes have no attendance policy at all.

Consequently:

- A missing attendance value might mean the student skipped class.
- Or the instructor forgot to upload records.
- Or attendance was never taken.

The spreadsheet simply contains **NA** but that **single symbol represents several very different realities**.

*Step 5: Institutional context*

Imagine two universities:

- **University A:** A student is labeled **dropout** after missing one semester.
- **University B:** Students **remain officially enrolled** for two years before being classified as dropouts.

Both datasets contain a variable called **dropout**, yet the labels represent **fundamentally different** institutional rules. Combining these datasets without understanding their origins would create misleading training data.

*The dataset is constructed*

Rather than viewing the dataset as objective truth, we can describe it as a unification of various items including **student behavior**, **university policies**, **measurement choices**, **sampling decisions**, and **data engineering**. The final spreadsheet reflects not only students' academic lives but also the university's **administrative procedures**, **technical systems**, and **operational definitions**.

*Evaluating fitness for purpose*

Now consider several possible uses of this same dataset.

- **Task 1, Annual reporting:** The university wants to know *What percentage of first-year students left last year?* The dataset is **probably sufficient**. Only approximate counts are needed.

- **Task 2, Early warning system:** The university wants to predict dropout six months before it happens. The same dataset may now be inadequate because:
  - Attendance data arrive weeks late.
  - Financial information is updated only once each semester.
  - Mental health indicators are absent.

  Although acceptable for historical reporting, the dataset lacks the timely information needed for real-time intervention.
- **Task 3, National AI model:** A government agency wants to train one model using this dataset for every university. This is problematic because:
  - Only one university contributed data.
  - Admissions policies differ elsewhere.
  - Student demographics differ.
  - Institutional **definitions of dropout** vary.

  The dataset is no longer representative of the intended population.
- **Task 4, Research on causes of dropout:** An education researcher asks: *Does library usage reduce dropout?* This dataset alone cannot answer that causal question. Students who use the library more may also be:
  - More motivated,
  - Better prepared academically,
  - Wealthier,
  - Supported by stronger social networks.

  Without **controlling** these **confounding factors**, the dataset may **reveal associations** but **not causal** relationships.

This example demonstrates both central ideas:

- *Data are produced, not simply found:* Every value in the dataset reflects **human decisions** about **definitions**, **measurements**, **sampling**, **recording**, and **institutional practices** rather than being a direct, objective capture of reality.
- *Data quality is purpose-dependent:* The same student dataset can be **entirely appropriate** for descriptive reporting, **yet inadequate** for real-time prediction, causal analysis, or deployment in different institutional

settings. In other words, a **dataset's quality is meaningful** only relative to a **specific task** and **context**, consistent with the formulation $\boldsymbol{Q(D, T, C)}$.

# Data Sources and Their Inherent Realities

We established that **data** is a **constructed artifact** rather than an **objective truth**. We now examine the **factories** where that construction takes place. The origin of a dataset is the **primary determinant** of its underlying **reality** and **reliability**. Whether data is **harvested** from human surveys, physical sensors, digital system logs, or bureaucratic administrative records, **each source** operates under a **fundamentally different data-generating process**. This process dictates the specific types of **errors**, **biases**, **governance requirements**, and **engineering challenges** that are inevitably baked into the data. Therefore, understanding the unique **mechanics**, **incentives**, and **limitations** of each data source is not merely a technical detail; it is a fundamental prerequisite for evaluating whether a dataset is **actually fit** for a specific operational purpose.

## 2.2.1 Surveys

Surveys collect information by **prompting respondents** with **structured** or **semi-structured questions** (e.g., household income, health questionnaires, customer satisfaction, public opinion polls).

- **Advantages:** Surveys can capture **latent constructs** that **rarely appear** in **routine databases**, such as *beliefs*, *preferences*, *intentions*, *self-reported behaviors*, *social circumstances*, and *perceived quality* of life.
- **Sources of Error**
  - **Coverage Error:** The sampling frame misses parts of the target population.
  - **Sampling Error:** The selected sample differs from the population purely by chance.
  - **Nonresponse Bias:** Respondents differ systematically from those who refuse to participate.
  - **Recall Bias:** Inaccurate memory of past events.
  - **Social-Desirability Bias:** Respondents provide answers they believe are socially acceptable rather than truthful.
  - **Instrument Effects:** Responses shift based on question wording, sequencing, or cultural/linguistic translation.

- **AI Implications:** A model trained on survey data may learn **response behavior** rather than the **intended construct** (e.g., self-reported exercise reflects both **actual physical activity** and the **respondent's willingness** to report it).
- **Required Documentation:** Target population, sampling frame, recruitment procedure, response rate, weighting schemes, exact questionnaire wording, administration mode, handling of missingness, language versions, and known limitations.

*Example # 39: Measuring physical activity through a health survey*

A national public health agency wants to estimate how physically active adults are and develop an AI model to identify individuals at risk of inactivity-related diseases.

*Step 1: Designing the survey*

Researchers create a questionnaire that asks: *During the past seven days, on how many days did you exercise for at least 30 minutes?* Additional questions collect:

- Age
- Gender
- Education
- Occupation
- Height and weight
- Smoking status
- Self-rated health

The target population is all adults aged 18 years and older.

*Step 2: Collecting responses*

The survey is distributed online. 10,000 people are invited; 6,000 complete the survey; 4,000 never respond. The resulting dataset **appears complete**, but several sources of error have already entered.

*Coverage error*

The survey is conducted **exclusively online**. People **without reliable internet** access – often **older adults** or those in **remote communities** – cannot participate. Although the target population is all adults, the actual dataset **overrepresents younger**, more **technologically connected** individuals.

*Sampling error*

Suppose the survey **randomly selects** more **residents** from **urban areas** than rural areas. By **chance**, the sample contains **more people with access to gyms** and **recreational facilities**. Consequently, the estimated average **exercise level** may be higher than that of the true population.

*Nonresponse bias*

Many people who are **physically inactive** choose **not to complete** the survey because they are **uninterested** in health research. As a result, **respondents** are generally **more health-conscious** than nonrespondents. The survey now **systematically overestimates** physical activity.

*Recall bias*

One respondent exercised only twice last week but **mistakenly remembers** exercising four times. Another **cannot accurately** remember how long each workout lasted. These errors arise from **imperfect memory** rather than **intentional deception**.

*Social-desirability bias*

A respondent knows that **exercising regularly** is **viewed positively**. Instead of reporting: *I exercised once* they answer: *I exercised five times*. The recorded value **reflects** not only **physical activity** but also the **respondent's desire** to **present** themselves **favorably**.

*Instrument effects*

Consider two versions of the question.

- **Version A**: *How many days did you exercise for at least 30 minutes last week?*
- **Version B**: *How many days did you engage in moderate or vigorous physical activity lasting at least 30 minutes during the previous seven days?*

Although both questions **aim to measure** the **same construct**, respondents **may interpret** *exercise* **differently** from *physical activity*. Someone who walks to work daily might answer 0 to **Version A** because they do not consider walking to be exercise but answer 5 to **Version B**. Small **wording changes** can therefore produce **systematically different responses**.

*AI implications*

Researchers now train an AI model to predict whether someone is physically active using these survey responses.

- However, the target variable is based on **self-reported exercise**, **not objectively measured** activity.
- The model may therefore learn patterns associated with **reporting behavior** rather than actual behavior.

For example, **individuals with higher education** may report their exercise routines **more accurately**, while **others** consistently **overestimate** their activity. The model could **incorrectly associate** education with physical activity simply because of differences in reporting accuracy. If the model is later applied to data collected from wearable fitness trackers, its **performance may deteriorate** because **wearable devices** measure **actual movement** rather than **self-reported behavior**.

*Why documentation matters*

To determine whether this survey dataset is suitable for research or AI development, future users need documentation such as:

- **Target population:** Adults aged 18 years and older.
- **Sampling frame:** Registered households with internet access.
- **Recruitment procedure:** Email invitations sent to randomly selected households.
- **Response rate:** 60% (6,000 of 10,000 invited participants).
- **Weighting scheme:** Responses weighted to match national age and sex distributions.
- **Questionnaire wording:** Exact text of every survey question.
- **Administration mode:** Self-administered online questionnaire.
- **Handling of missing data:** Incomplete responses excluded or imputed using specified methods.

- **Language versions:** English, Spanish, and French translations.
- **Known limitations:** Potential coverage error, nonresponse bias, recall bias, and social-desirability bias.

A survey dataset **does not provide** a **direct measurement of reality**. Instead, it reflects a **combination** of the **underlying phenomenon** (physical activity), **respondents' memory** and **willingness to disclose** information, the survey's wording and administration, and the sampling process. Consequently, an AI model trained on survey data may learn **how people answer questions** as much as **the real-world behavior the survey intends to measure**. Proper documentation is therefore essential for assessing whether the data are fit for a particular purpose.

### 2.2.2 Sensors

Sensors **convert physical phenomena** into recorded **digital signals** (e.g., temperature probes, accelerometers, cameras, microphones, wearables, ECG systems, industrial vibration sensors, weather stations, satellites, vehicle telemetry).

- **Measurement model:** A basic sensor measurement can be represented as: $X_{\text{observed}} = X_{\text{true}} + b + \epsilon$, *where* $X_{\text{true}}$ is the underlying quantity, $b$ is systematic bias, and $\epsilon$ is random measurement noise.

  More realistically, it is a complex, non-linear function:

  $$X_{\text{observed}} = g(X_{\text{true}}, Z, \theta_{\text{device}}) + \epsilon$$

  where $Z$ represents environmental conditions and $\theta_{\text{device}}$ describes device-specific behavior.
- **Sensor-Data Challenges:** Calibration drift, device replacement, firmware updates, irregular sampling, clock misalignment, packet loss, signal saturation/clipping, environmental interference, changing units, delayed transmission, and duplicated events.
- **Critical questions:** Was the sensor calibrated? What is its measurement range and temporal resolution? How are failures encoded? Does a *zero* mean zero or missing? Are timestamps based on device time or server time? Did hardware change during collection? Are measurements comparable across different sites?

- **Critical questions:** Was the sensor calibrated? What is its measurement range and temporal resolution? How are failures encoded? Does a *zero* mean zero or missing? Are timestamps based on device time or server time? Did hardware change during collection? Are measurements comparable across different sites?
- **The core takeaway:** Sensor data may appear highly objective, but the physical instrument and the acquisition process fundamentally shape what is observed.

### 2.2.3 Web data

Web data is **harvested** from websites, social media platforms, search queries, online forums, product reviews, news sources, digital advertisements, and public APIs[197].

- **Advantages:** Massive volume, rapid availability, diverse content, naturalistic behavior, and broad topical coverage.
- **Limitations:** Platform users **rarely represent** the **general population**; platform algorithms **actively influence** what users **see and produce**; bots and coordinated campaigns **distort observations**; content is frequently **duplicated**; user identities are **uncertain**; interfaces change constantly; collection may **violate Terms of Service**, copyright, or privacy expectations; deleted content may persist in archives; and demographic attributes are often **inferred unreliably**.
- **The platform as measurement:** Observed web behavior is **not a direct measure** of stable human preference. It is **a reaction** within an **engineered environment**, defined by a combination of **user**, **interface**, **ranking algorithm**, **moderation**, and **platform incentives**.

*Example # 40: Predicting public opinion from social media posts*

Suppose a government agency wants to **estimate public support** for a **new environmental policy**. Instead of conducting a nationwide survey, analysts

[197] An **API (Application Programming Interface)** is a set of rules and endpoints that enables different software applications to communicate, share data, and use each other's functionality without exposing their internal implementation. For example, a weather app uses the OpenWeather API to retrieve real-time weather information, an e-commerce website uses the Stripe API to process online payments, a travel app uses the Google Maps API to display maps and directions, and a music application uses the Spotify API to access songs, playlists, and user data.

collect **5 million public posts** from a popular social media platform and train an AI model to classify each post as **supportive**, **neutral**, or **opposed**.

At first glance, the dataset **appears ideal**: it is **enormous**, **continuously updated**, and contains **spontaneous public discussions**. However, several limitations emerge.

*Non-representative users*

- Only a **fraction of the country's population** actively uses the platform.
- **Older** adults, people living in **rural areas**, and individuals with limited internet access rarely post online, while **younger urban users** are highly active.

Although the AI model accurately summarizes opinions expressed on the platform, it **cannot be assumed** to represent the **opinions of the entire population**.

*Platform algorithms shape the data*

- The platform's recommendation algorithm **promotes posts** that generate **high engagement**.
- As **controversial** or **emotionally charged** posts receive **more likes**, comments, and shares, they become **increasingly visible**.

Consequently, the dataset contains a **disproportionately large number** of **polarized opinions**, while **moderate viewpoints** receive much **less exposure** and are therefore less likely to be collected. The observed data reflect both **public opinion** and **the platform's ranking algorithm**.

*Bots and coordinated campaigns*

During the policy debate, **hundreds of automated** accounts **repeatedly post identical messages** supporting one side of the issue. Without detecting and **removing these accounts**, the AI model **may interpret** coordinated automation as genuine public sentiment.

*Duplicate content*

- A popular post criticizing the policy is shared 100,000 times.

- If every repost is treated **as an independent observation**, one individual's opinion effectively appears 100,001 times in the dataset.

The measured **frequency of opinions** becomes **inflated by duplication** rather than reflecting the number of unique individuals.

*Uncertain user identity*

A user's profile indicates they are a 30-year-old resident of the capital city. In reality, the account belongs to:

- a teenager,
- a company managing multiple social media accounts,
- or an anonymous individual using false profile information.

Any demographic analyses based on user profiles therefore contain **considerable uncertainty**.

*Changing platform interfaces*

- Halfway through data collection, the platform **introduces a new recommendation system** that strongly **favors short video** posts over text.
- The number of text posts discussing the policy immediately declines; not necessarily because public interest decreased, but **because the platform changed** what users were encouraged to create.

A sudden shift in the dataset may therefore **reflect a platform redesign** rather than a real change in public opinion.

*Legal and ethical constraints*

Researchers collect posts using an unofficial web scraper. However:

- the platform's Terms of Service prohibit automated scraping,
- some posts contain copyrighted material,
- and users who later delete their posts discover that archived copies remain in the research dataset.

Even if technically feasible, data collection **may raise legal and ethical concerns** regarding **consent**, **copyright**, and **privacy**.

*The platform as the measurement instrument*

Unlike a traditional survey, where researchers design the questionnaire, a **social media platform** is itself the **measurement instrument**. A user's observed behavior depends **not only on** their **personal opinion** but also on:

- what the recommendation algorithm shows them,
- which posts moderators remove,
- which topics are currently trending,
- how the interface encourages interaction,
- and the social incentives created by likes, shares, and comments.

Thus, a social media post is **not simply a direct expression** of an individual's stable beliefs; it is a response produced within a **carefully engineered digital environment**.

*AI implications*

Researchers train an AI model to predict public opinion from these posts. The model achieves excellent accuracy on the social media dataset. However, **when applied to responses** from a **nationally representative survey**, its performance **drops substantially**. This is because the model has learned patterns that are partly specific to the platform, including:

- the language commonly used by social media users,
- engagement-driven posting behavior,
- algorithmically amplified topics,
- coordinated bot activity,
- and platform-specific communication styles.

Rather than learning **general public opinion**, the model has learned **how people express opinions within that particular online ecosystem**.

*Why Documentation Matters*

Anyone using this dataset should know:

- **Source platforms:** Which websites or social media platforms were used?
- **Collection method:** Official API, web scraping, or third-party archive?
- **Collection period:** When were the data collected?
- **Sampling strategy:** Were all posts collected, or only those matching certain keywords or hashtags?

- **Preprocessing:** How were duplicate posts, spam, bots, and deleted content handled?
- **Identity limitations:** Were demographic characteristics verified or merely inferred?
- **Legal and ethical considerations:** Did data collection comply with platform policies, copyright law, and privacy regulations?
- **Known biases:** How might recommendation algorithms, moderation policies, or platform demographics affect the dataset?

Web data provide an **unprecedented view of online behavior**, but they do not offer a direct or unbiased measurement of society. Every observation is shaped by the interaction between **users**, **platform algorithms**, **interface design**, **moderation policies**, and **platform incentives**. Consequently, AI models trained on web data often learn patterns of **platform-specific behavior** as much as they learn the underlying human attitudes or preferences they are intended to measure.

### 2.2.4 System and application logs

Logs are **automated records** produced by **software systems**, **applications**, **networks**, and **devices** (e.g., website clickstreams, application events, server errors, authentication logs, model-inference logs, database queries, transaction events).

- **Strengths:** They provide **detailed event sequences**, high temporal resolution, operational evidence, large-scale behavioral traces, and critical support for monitoring and debugging.
- **Limitations:** A logged event **only proves** that software recorded something – *not necessarily* that the real-world event happened exactly as interpreted. **Common problems** include inconsistent event definitions, missing instrumentation, schema changes, duplicated events after network retries, out-of-order arrival, client-side blockers (e.g., ad blockers), clock differences, silent logging failures, and unintended changes introduced by application updates.
- **Event Semantics:** For every logged event, you must rigorously distinguish between:

- **Event Time:** When the real-world action actually occurred.
- **Recording Time:** When the local system logged it.
- **Arrival Time:** When it reached the central data pipeline.
- **Processing Time:** When the pipeline actually handled it.

Note: These timestamps are a **primary cause** of **temporal data leakage** and **incorrect feature windows**.

### 2.2.5 Administrative records

Administrative records are generated to **support organizational operations**, not scientific research (e.g., billing data, hospital records, school enrollment, tax records, insurance claims, police/court records, employment records, social-service databases).

- **Advantages:** They **cover large populations**, offer **longitudinal tracking**, have **high operational relevance**, incur **low marginal collection costs**, and connect directly to institutional decisions.
- **Limitations:** Administrative data encodes **institutional behavior** not objective reality. For example:
  - A diagnosis code may reflect billing optimization, not clinical severity.
  - An arrest record reflects policing patterns and enforcement priorities, not the true rate of offending.
  - Healthcare expenditure reflects access to care and treatment protocols, not just medical need.
  - School absences may reflect transportation or caregiving constraints, not truancy.
  - Unemployment records entirely omit people outside the formal labor system.
- **The core formulation:** Observed outcome is a function of Underlying Condition, Access, Policy, and Institutional Practice. Administrative fields must be carefully interpreted and never blindly accepted as ground truth.

*Example # 41: Predicting hospital readmissions from administrative records*

A national health service wants to develop an AI model that predicts which patients are likely to be readmitted to the hospital within 30 days after

discharge. Rather than collecting new clinical measurements, researchers use existing **hospital administrative records**, including admission dates, diagnosis codes, procedure codes, insurance claims, length of stay, and billing information.

The **dataset is attractive** because it covers millions of patients over many years. However, the records were created to **support hospital operations** and **reimbursement** – **not to measure** patients' true health status.

*Billing codes versus clinical reality*

Two patients both have diabetes: **Patient A** has mild, well-controlled diabetes and **Patient B** has severe diabetes with multiple complications. Suppose **Hospital X** carefully records every complication because **doing so increases reimbursement**, whereas **Hospital Y** records only the **primary diagnosis**.

Although **Patient B** is much sicker, both patients may receive the same diagnosis code. Conversely, hospitals with **more detailed coding practices** may appear to treat sicker patients simply because they document conditions more thoroughly. The diagnosis code therefore reflects both **clinical condition** and **billing practice**.

*Access to care influences the data*

Consider two patients who develop the same post-surgical complication.

- **Patient A** lives near a hospital and is readmitted within two days.
- **Patient B** lives in a remote area with limited transportation and never returns to the hospital.

The administrative records show only one readmission. The **absence of a readmission record** does not necessarily mean Patient B recovered successfully; it **may simply indicate** limited access to healthcare.

*Institutional policies shape the records*

- **Hospital A** has a policy of admitting patients for overnight observation.
- **Hospital B** treats similar patients in an outpatient clinic without formal admission.

Both hospitals provide appropriate medical care, but **only Hospital A** generates hospital admission records. Differences in administrative policies **create differences in the dataset** even when patients have **similar medical conditions**.

*Missing populations*

The dataset contains records only for patients treated within the public hospital system. It excludes:

- people treated exclusively in private hospitals,
- individuals who never seek medical care,
- uninsured patients receiving informal care,
- and people who die before reaching a hospital.

The dataset therefore represents **healthcare utilization**, not the complete burden of disease in the population.

*The AI model learns institutional patterns*

Researchers train an AI model to predict readmissions. The model discovers that patients from **certain neighborhoods** have **higher readmission rates**. It would be tempting to conclude that these neighborhoods contain sicker patients. However, alternative explanations exist. Residents may:

- have better access to hospitals,
- live closer to emergency departments,
- receive more frequent follow-up care,
- or be treated by hospitals with lower thresholds for readmission.

The model is therefore learning patterns of **healthcare delivery** as well as patterns of **patient health**.

*Understanding the observed outcome*

A hospital readmission is not determined solely by a patient's medical condition. Instead, it results from several interacting factors including **underlying health condition**, **access to healthcare**, **hospital policies**, and **administrative practices**. Consequently, a readmission record should not be interpreted as a direct measure of illness severity.

*Why Documentation Matters*

Anyone using these administrative records should know:

- **Purpose of collection:** Hospital administration, billing, and reimbursement.
- **Population covered:** Patients treated within participating hospitals.
- **Coding standards:** Which diagnosis and procedure coding systems were used?
- **Policy context:** Were reimbursement rules or admission policies changed during the study period?
- **Data linkage:** How were repeated visits by the same patient identified?
- **Known limitations:** Differences in coding practices, healthcare access, institutional policies, and missing populations.

### 2.2.6 Controlled experiments

Controlled experiments **deliberately assign** or **manipulate conditions** to estimate causal effects. In a **randomized experiment**, treatment assignment is modeled as $T_i \sim \text{Bernoulli}(p)$, intended to **make treatment groups comparable** in expectation and support causal inference.

- **Strengths:** Clearer intervention definition, strict temporal ordering, reduced confounding, prespecified outcomes, and controlled measurement procedures.
- **Limitations:** Ethical or practical constraints, participant noncompliance, dropout/attrition[198], contamination between treatment groups, limited external validity (generalizability), imperfect outcome measurement, selective recruitment, and implementation differences across sites.
- **The core takeaway:** Experimental data is not automatically unbiased. Study design, adherence rates, missing outcomes, and generalizability still require rigorous scrutiny.

### 2.2.7 Public datasets

[198] **Attrition** is the loss of participants from a research study before it is completed, which may reduce the study's statistical power and introduce bias if the participants who withdraw differ systematically from those who remain. For example, in a 12-month clinical trial, some participants may drop out due to side effects, relocation, personal commitments, or loss of interest, resulting in a smaller final sample and potentially affecting the validity and reliability of the study's findings.

Public datasets are released for **research**, **education**, **benchmarking**, or **policy use** (e.g., government open-data portals, scientific repositories, ML benchmarks, satellite archives, demographic surveys, medical-image challenges).

- **Advantages:** High accessibility, reproducibility, shared benchmarks, lower entry barriers, standardized method comparison, and educational value.
- **Risks:** Outdated versions, unclear provenance[199], inappropriate reuse, hidden preprocessing steps, label errors, undocumented exclusions, benchmark saturation (models overfitting to the test set), privacy risks, and license restrictions.
- **The core warning:** A dataset being *public* **does not mean** it is **free of ethical concerns**, that **every use is permitted**, that its labels are correct, that it represents a real deployment setting, or that original consent covered all possible future uses.
    - *Evidence:* Northcutt, Athalye, and Mueller (2021)[200] identified pervasive label errors in the test sets of ten widely used computer vision, language, and audio benchmarks, proving that even prestigious public datasets contain enough noise to invalidate model comparisons.

*Example # 42: Pervasive Label Errors in Test Sets*

Northcutt, Athalye, and Mueller showed in their study that:

- **Label errors are widespread** in the test sets of many widely used machine learning benchmarks.
- **Test set labels are often assumed to be correct**, but the authors found that many contain incorrect or ambiguous annotations.
- **Errors exist across multiple domains**, including:
    - Image classification
    - Natural language processing (NLP)

[199] **Provenance** refers to the **documented origin**, **history**, and **chain of custody** of a dataset, including details about how, when, where, and by whom the data were collected, processed, transformed, and maintained. **Clear data provenance** helps researchers assess the **reliability**, **quality**, and **suitability of a dataset** for a particular study while promoting **transparency** and **reproducibility**. For example, a medical dataset with well-documented provenance includes information about the hospitals where the data were collected, the time period of collection, preprocessing and anonymization procedures, and any modifications made before the dataset was released for research use.

[200] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller, *Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks*, NeurIPS, 2021.

    - Named entity recognition
    - Sentiment analysis
    - Question answering
- **Incorrect test labels distort benchmark results**, making model performance appear better or worse than it actually is.
- **Small differences between state-of-the-art models may not be meaningful** because they are often smaller than the number of mislabeled test examples.
- **Benchmark rankings can change** after correcting mislabeled test samples, meaning the *best* model under the original benchmark may no longer be the best.
- The authors introduced **Confident Learning**, a method for identifying likely label errors by estimating joint distributions between noisy and true labels.
- **Human reviewers verified** many of the suspected label errors identified by Confident Learning, confirming that the method is highly effective.
- The study corrected **thousands of mislabeled examples** across numerous popular benchmark datasets.
- **Benchmark instability** caused by label errors reduces the reliability of reported model improvements.
- The paper argues that **higher model accuracy does not necessarily reflect genuine progress** if evaluation datasets contain significant annotation mistakes.
- The authors recommend that benchmark datasets undergo **regular auditing and label correction** to maintain their reliability.
- They advocate for **continuous dataset maintenance** rather than treating benchmark datasets as fixed and error-free.
- The work highlights that **data quality is as important as model quality** for advancing machine learning research.
- Overall, the study concludes that **improving benchmark labels leads to more trustworthy model evaluation, fairer comparisons, and more reliable scientific progress**.

### 2.2.8 Synthetic Data

Synthetic data consists of **artificially generated observations** designed to **mimic the statistical properties** of real data. It is produced by rule-based simulators[201], mechanistic models[202], agent-based models[203], GANs[204], VAEs[205], diffusion models[206], LLMs[207], or statistical sampling from fitted distributions[208].

- **Potential Uses:** Simulating rare events, privacy-preserving data access, augmenting underrepresented classes, testing data pipelines, training in unsafe/expensive environments, generating counterfactual scenarios, and balancing datasets
- **Advantages:** Offers controlled labels, highly configurable scenarios, and reduces dependence on direct access to sensitive, regulated records.
- **Risks:** Synthetic data **can reproduce biases** from the source data, **omit rare subgroups**, **oversimplify real-world variability**, introduce **unrealistic correlations**, **leak information** from original training records, create **misleadingly clean labels**, and encourage **unwarranted confidence** in simulation-to-reality transfer. (*Note: Synthetic does not mean anonymous or risk-free; generated health data often retains residual disclosure risks*).

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[201] **Rule-based simulators:** Computer programs that generate synthetic data by following predefined rules and logical conditions that mimic real-world systems. Example: A traffic simulator that models vehicle movement using fixed traffic laws and signal timings.

[202] **Mechanistic models:** Mathematical or computational models that simulate a system based on the underlying physical, biological, or chemical mechanisms governing its behavior. Example: A climate model that predicts temperature changes using atmospheric physics equations.

[203] **Agent-based models (ABMs):** Simulation models in which autonomous agents interact according to defined behavioral rules, producing complex system-level outcomes. Example: An epidemic simulation where individuals (agents) move, interact, and spread an infectious disease.

[204] **GANs (Generative Adversarial Networks):** Deep learning models that generate realistic synthetic data by training two neural networks – a generator and a discriminator – in competition with each other. Example: Creating realistic human face images that do not correspond to real people.

[205] **VAEs (Variational Autoencoders):** Probabilistic deep learning models that learn compressed representations of data and generate new samples by decoding points from a learned latent space. Example: Generating new handwritten digits resembling those in the MNIST dataset.

[206] **Diffusion models:** Generative AI models that create new data by learning to reverse a process that gradually adds noise to training data until a clean sample is reconstructed. Example: AI systems that generate photorealistic images from text prompts.

[207] **LLMs (Large Language Models):** Neural network models trained on massive text corpora to understand and generate human-like language. Example: ChatGPT generating synthetic conversations, summaries, or programming code.

[208] **Statistical sampling from fitted distributions:** A technique that generates synthetic data by randomly drawing samples from a probability distribution estimated from observed data. Example: Fitting a normal distribution to patient blood pressure measurements and sampling new values to create a synthetic dataset.

- **Validation Dimensions:** A synthetic dataset must be rigorously evaluated for:
  - **Fidelity:** Does it preserve relevant distributions and relationships?
  - **Utility:** Does it actually support the intended analysis or model training?
  - **Privacy:** Can original individuals be inferred or reconstructed?
  - **Fairness:** Are subgroup properties preserved appropriately without amplifying bias?
  - **Coverage:** Are rare but critical edge cases represented?
  - **Realism:** Do the generated combinations obey real-world domain constraints?

*Example # 43: Synthetic data for hospital readmission prediction*

A hospital wants to develop a machine learning model that predicts whether patients are likely to be readmitted within 30 days. Due to **strict privacy** regulations, researchers **cannot freely access real patient records**.

Instead, they create a **synthetic dataset** using a generative model **trained** on **historical hospital data**. The synthetic records resemble real patients statistically but do not directly correspond to actual individuals (**Table 5**).

**Table 5: Synthetic patient-level data for illustrating 30-day hospital readmission prediction.** The table presents five hypothetical patient records containing age, diabetes status, length of stay, number of previous admissions, and 30-day readmission outcome. The data are entirely synthetic and are intended solely to demonstrate the structure of a simple supervised-learning dataset for a binary clinical prediction task.

| Patient ID | Age | Diabetes | Length of Stay (days) | Previous Admissions | Readmitted (30 days) |
|---|---|---|---|---|---|
| **S001** | 72 | Yes | 8 | 3 | Yes |
| **S002** | 45 | No | 2 | 0 | No |
| **S003** | 67 | Yes | 5 | 1 | Yes |
| **S004** | 31 | No | 1 | 0 | No |
| **S005** | 81 | Yes | 10 | 4 | Yes |

*The table above is entirely synthetic and is used only for illustration.*

*How this synthetic data is used*

- **Privacy-preserving research:** Data scientists develop prediction models without accessing identifiable patient records.
- **Balancing the dataset:** Additional synthetic patients who experienced rare complications are generated so the model learns these cases.
- **Testing software:** Developers verify that hospital analytics systems work correctly before deployment.
- **Counterfactual analysis:** Researchers simulate scenarios such as *What if average hospital stay were reduced by two days?*

*Advantages demonstrated*

- No direct use of identifiable patient records during model development.
- Researchers can generate thousands of additional examples of uncommon medical conditions.
- Labels (e.g., whether a patient is readmitted) are known and can be controlled.
- Different hospital scenarios can be simulated quickly and inexpensively.

*Potential risks*

Suppose the original hospital data **underrepresented younger patients** with **chronic illnesses**. The synthetic generator may **inherit this bias**, producing too few such cases. As a result:

- The prediction model performs well for older adults but **poorly for younger** patients.
- **Rare disease** combinations may be **missing**.
- If the generator **memorized parts** of the training data, some synthetic records might **reveal information** about real patients.
- The synthetic records may appear **unrealistically clean**, lacking missing values or recording errors that occur in real hospitals.

*Validation Checklist*

Before using the synthetic dataset, researchers evaluate several dimensions depicted in **Table 6**. Synthetic data can substantially improve privacy, experimentation, and model development, but it is **not automatically safe or representative**. A synthetic dataset is only valuable if it is carefully validated

for **fidelity, utility, privacy, fairness, coverage, and realism** before being used for analysis or machine learning.

**Table 6: Validation dimensions for synthetic clinical data.**

| Validation Dimension | Example Check |
|---|---|
| **Fidelity** | Do age distributions, disease frequencies, and correlations resemble those in the real data? |
| **Utility** | Does a model trained on synthetic data achieve similar performance when evaluated on real data? |
| **Privacy** | Can an attacker infer whether a real patient's record was used to generate the synthetic dataset? |
| **Fairness** | Are outcomes equally representative across age groups, genders, and ethnicities? |
| **Coverage** | Are rare but clinically important cases (e.g., severe sepsis, uncommon drug reactions) sufficiently represented? |
| **Realism** | Are impossible combinations (e.g., a male patient marked as pregnant) absent from the generated data? |

# Data Quality

**High-quality data** are the foundation of **reliable statistical analysis**, scientific **research**, and **trustworthy** artificial intelligence. However, data quality is *not a single property* that can be summarized by one metric. Instead, it **encompasses multiple dimensions** that describe different aspects of how well a dataset supports its intended use. A dataset may contain **complete** records yet be **inaccurate**, **timely** but **unrepresentative**, or internally **consistent** while suffering from **systematic measurement error**. Consequently, evaluating data quality requires a multidimensional perspective that considers **several complementary characteristics**. This module examines the principal dimensions of data quality – including completeness, accuracy, consistency, timeliness, representativeness, and measurement error – and explains how each contributes to the **reliability**, **validity**, and **practical usefulness** of data for analysis and decision-making.

## 2.3.1 Completeness

Completeness measures whether **all required data** is **actually present**. It can be evaluated at different granularities:

- **Cell-Level Completeness:** $\text{Completeness}_{\text{j}} = 1 - \left(\frac{m_j}{n}\right)$, where $m_j$ is the number of missing cell values in variable $j$ out of $n$ total records.
- **Record-Level Completeness:** The proportion of total records that contain every single required field i.e.,

$$\text{Completeness}_{\text{record}} = \frac{\text{records containing all required fields}}{\text{total records}}$$

### Types of Incompleteness

Incomplete data can arise in several forms, including:

- missing cells or attribute values;
- missing records;
- missing time intervals;
- missing population subgroups;
- missing data modalities;
- missing labels;
- missing metadata;

- and missing provenance information.

For example, a dataset may have 99% of its cells populated while completely excluding an important or vulnerable population.

#### Context matters

Missing information is not always a defect.

- A laboratory test may be absent because it was **clinically unnecessary**.
- A survey respondent may **intentionally skip a question**.
- A sensor reading may be **unavailable** because **no sensor was installed** at a particular location.

The critical question is therefore: **Does the missing information prevent reliable use of the dataset for its intended purpose?**

### 2.3.2 Accuracy

Accuracy measures **how closely** recorded values correspond to the **true** or **accepted reference** values.

- **For numerical measurements:** The error for a single observation is computed as follows: $e_i = x_i - x_i^*$ where $x_i^*$ is the trusted reference value.

  This is often summarized using the Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n}\sum |x_i - x_i^*|$$

- **For categorical data:** $\text{Label Accuracy} = \frac{\text{correctly labeled cases}}{\text{reviewed cases}}$

#### Sources of inaccuracy

Common sources of inaccurate data include:

- manual data-entry errors;
- faulty or poorly calibrated sensors;
- incorrect record linkage;
- software or coding errors;
- ambiguous variable definitions;
- outdated information;
- disagreement among annotators;
- transcription mistakes;

- and data conversion or formatting errors.

Accuracy **must always be assessed** against an **independent** and **trustworthy reference**. Agreement within a database does not, by itself, establish correctness.

### 2.3.3 Consistency

Consistency asks whether data values **agree logically** across **formats**, **records**, **systems**, and **established rules**.

- **Formatting:** "Male", "M", "1", and "man" representing the exact same category; dates stored as DD/MM/YYYY in one column and MM/DD/YYYY in another.
- **Logical Impossibilities:** A patient recorded as deceased prior to a later clinical visit; a child's age recorded as older than their parent's age.
- **Structural:** Different units of measurement mixed in the same column; conflicting labels for the same entity across different files; duplicate IDs assigned to different people.

#### Constraint-based consistency

Many inconsistencies violate strict mathematical or domain constraints, such as:

- $\text{discharge data} \geq \text{admission date}$
- $0 \leq \text{probability} \leq 1$
- $\sum_{k=1}^{K} p_k = 1$ (probabilities must sum to one)

While **constraint violations** can often be **detected automatically** via data validation scripts, resolving the underlying conflict **almost always requires domain expertise**.

### 2.3.4 Timeliness

Timeliness concerns whether data is **sufficiently current** and **available exactly when it is needed** for a decision. It is broken down into four distinct concepts:

- **Freshness:** How old the underlying information actually is.

- **Latency:** The delay between the real-world event occurring and the data becoming available in the system.
- **Update frequency:** How often the values are refreshed.
- **Validity period:** How long a specific data point remains operationally relevant.

A real-time fraud system may **require** data **within milliseconds**, whereas annual demographic planning **may tolerate months** of delay.

#### Staleness

At prediction time $t$, the age of a feature can be expressed as

$$\Delta t = t - t_{\text{last update}}$$

If $\Delta t$ **exceeds a task-specific threshold**, the feature should be **considered stale** and may reduce model performance.

#### Timeliness and leakage

Historical datasets typically record **when an event occurred**, whereas operational systems require knowledge of **when the information became available**.

Using **event timestamps** instead of **data-availability timestamps** may unintentionally provide models with information that would not have been available at prediction time, resulting in **overly optimistic** performance estimates through **data leakage**.

### 2.3.5 Representativeness

Representativeness asks whether the data **adequately reflects the intended target population** and the **actual operating conditions** of deployment.

Let $P_{\text{data}}(X, Y)$ denote the **dataset distribution** and $P_{\text{target}}(X, Y)$ the **deployment distribution**. A representativeness problem arises when:

$$P_{\text{data}}(X, Y) \neq P_{\text{target}}(X, Y)$$

in **consequential ways** i.e., ways that materially affect the model's performance, calibration, fairness, robustness, or safety when deployed.

- **Performance degradation**

- The model encounters feature values or combinations that were rare or absent during training.
- Example: A speech recognition system trained mostly on American English performs poorly on Scottish accents.

- **Unequal performance across subgroups**
  - Certain demographic or operational groups are underrepresented in the training data.
  - Example: A facial recognition model trained primarily on light-skinned faces has substantially lower accuracy on darker-skinned faces.
- **Different class proportions (label distribution shift)**
  - The frequency of outcomes differs between training and deployment.
  - Example: A fraud detection model trained when fraud represented 1% of transactions is deployed where fraud constitutes 10%, leading to poorly calibrated predictions.
- **Different feature distributions (covariate shift)**
  - The input variables have changed.
  - Example: A medical diagnosis model trained using data from one hospital is deployed in another hospital where patients have different demographic characteristics or clinical practices.
- **Changed relationships between inputs and outputs (concept shift)**
  - The mapping from $X$ to $Y$ changes.
  - Example: Consumer purchasing behavior changes after a major economic recession, making historical prediction models less accurate.
- **Missing deployment scenarios**
  - Important operating conditions are absent from the training data.
  - Example: An autonomous driving model trained only in daylight performs poorly at night or in heavy snow.

Data can be unrepresentative across multiple axes: demographic, geographic, socioeconomic, linguistic, clinical, temporal, device-related, environmental, or behavioral.

### Representativeness versus sample size

**A large dataset is not necessarily representative.** Collecting millions of additional observations from an already well-represented group does not compensate for the absence of data from underrepresented populations.

#### Intersectional evaluation

Evaluating representativeness across single dimensions (e.g., looking at gender and skin type separately) is often insufficient. Model performance **frequently degrades worst** at the intersection of **multiple marginalized attributes**. The landmark **Gender Shades**[209] study proved this by demonstrating that commercial facial recognition systems, which performed well on overall gender and overall skin type, **failed catastrophically** when evaluated at the intersection of darker skin and female gender.

### 2.3.6 Measurement Error

Measurement error is the **fundamental difference** between an **observed value** and the **true underlying construct** or **quantity** it is intended to measure.

- **Random error:** $X^* = X + \epsilon$ where $\mathbb{E}[\epsilon] = 0$.
  - Random noise **increases variance**, **weakens statistical associations**, and **reduces model precision**.
  - It **does not** inherently bias the central estimate.
- **Systematic error:** $X^* = X + b + \epsilon$ where $b \neq 0$.
  - A systematic bias creates a **persistent**, **directional shift** in the data, leading to **fundamentally flawed** conclusions.
- **Differential Measurement Error:** Error rates may differ significantly across subgroups.
  - $P(X^* \neq X | A = a)$ may vary depending on a sensitive attribute $A$.
  - This is a critical issue in fairness analysis.
  - If a model is trained on data where **one demographic group** is measured with **significantly more error** than another, the model **cannot mathematically correct** for those **underlying systemic deficiencies**.

[209] Joy Buolamwini and Timnit Gebru, *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*, *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, PMLR 81:77-91, 2018.

### Construct Validity (The Ultimate Error)

The most dangerous measurement error occurs before the instrument is even used: **the chosen variable simply does not measure the intended concept**.

- Using **healthcare cost** as a proxy for **health need**.
- Using **arrest records** as a proxy for **actual crime commission**.
- Using **user clicks** as a proxy for **genuine satisfaction**.
- Using **standardized exam performance** as a proxy for **innate intelligence**.
- Using **school absence** as a proxy for **student disengagement**.

### The core takeaway

If **proxy variable** is a **poor reflection** of the **real-world construct** we actually care about, **no amount of algorithmic tuning** will save the model. We will simply build a highly optimized system for **measuring the wrong thing**.

*Example # 44: Predicting hospital readmission*

Suppose a hospital wants to build a machine learning model that predicts whether a patient will be readmitted within 30 days after discharge. **Table 7** displays the items in the record. This dataset appears perfectly usable; but each data quality dimension can still fail independently.

**Table 7: Illustrative patient-level data for 30-day hospital readmission prediction.**

| Patient ID | Age | Sex | Blood Pressure | Diabetes | Discharge Date | Readmitted (30 days) |
|---|---|---|---|---|---|---|
| P001 | 72 | F | 145 | Yes | 2026-01-12 | Yes |
| P002 | 56 | M | 118 | No | 2026-01-13 | No |
| P003 | 81 | F | 160 | Yes | 2026-01-13 | Yes |

*The multidimensional nature of data quality*

Imagine six different versions of the exact same dataset (**Table 8**). Each dataset has **exactly one problem**, illustrating that **data quality is not a single number**. For example,

**Table 8: Comparative data-quality assessment across six datasets.**

| Dataset | Completeness | Accuracy | Consistency | Timeliness | Representativeness | Measurement Error |
|---|---|---|---|---|---|---|
| A | ✓ | X | ✓ | ✓ | ✓ | ✓ |
| B | ✓ | ✓ | ✓ | X | ✓ | ✓ |
| C | ✓ | ✓ | ✓ | ✓ | X | ✓ |
| D | X | ✓ | ✓ | ✓ | ✓ | ✓ |
| E | ✓ | ✓ | X | ✓ | ✓ | ✓ |
| F | ✓ | ✓ | ✓ | ✓ | ✓ | X |

- Dataset A records every patient's blood pressure, but every value is shifted upward by 20 mmHg because the **monitor is miscalibrated**.
- Dataset D has perfectly measured blood pressure, but 25% of patients have **no blood pressure recorded**.
- Dataset C contains accurate measurements, but almost **all patients are elderly**, making it poor for predicting outcomes in younger adults.

Each dataset is *good* in some respects and *bad* in others.

*Completeness*

Suppose the blood pressure column contains missing values (Table 9).

**Table 9: Illustrative patient blood-pressure data with missing observations.**

| Patient | Blood Pressure |
|---|---|
| P001 | 145 |
| P002 | NA |
| P003 | 160 |
| P004 | NA |
| P005 | 130 |

- There are 5 patients and 2 missing values.
- Cell-level completeness is $\left(1-\frac{2}{5}\right)=0.60$. The data is 60% complete.

Each dataset is *good* in some respects and *bad* in others.

Suppose every patient should have age, gender, blood pressure, and diabetes status. **Table 10** shows a record sample.

**Table 10: Record-completeness status for illustrative patient entries.**

| Patient | Complete |
|---|---|
| P001 | ✓ |
| P002 | X |
| P003 | ✓ |
| P004 | X |
| P005 | ✓ |

- Only 3 out of 5 records are complete.
- Record completeness equals $\frac{3}{5}=60\%$.

Completeness is **dependent on context**:

- Suppose a blood test is **only ordered** for diabetic patients.
- Missing blood pressure for a **non-diabetic** patient may be perfectly acceptable.
- The missing value is **not a data quality problem** because it was never clinically required.

*Accuracy*

Suppose the patient's true blood pressure is 140 mmHg but the dataset records 147 mmHg. The measurement error is $e=147-140=7$ mmHg. Now consider five patients (**Table 11**). The mean absolute error is $\frac{7+2+3+5+2}{5}=3.8$ mmHg. Thus, measurements are off by **3.8 mmHg on average**.

Table 11: Illustration of absolute measurement error.

| True | Recorded | Absolute Error |
|---|---|---|
| 140 | 147 | 7 |
| 120 | 118 | 2 |
| 155 | 152 | 3 |
| 130 | 135 | 5 |
| 160 | 158 | 2 |

*Label accuracy*

Suppose experts review 100 readmission labels. Only 95 are correct. Label accuracy is $\frac{95}{100} = 95\%$. Notice that **every value is present (complete)**, but **5% are wrong (inaccurate)**.

*Consistency*

Table 12 displays the diabetes column in a dataset. Every value means exactly the same thing. The data is accurate but inconsistent. A computer may interpret these as **five different categories**.

Table 12: Illustrative inconsistency in categorical data encoding.

| Patient | Diabetes |
|---|---|
| P001 | Yes |
| P002 | Y |
| P003 | 1 |
| P004 | True |
| P005 | Diabetic |

*Logical inconsistency*

If the discharge date of a patient is $\text{Jan } 10$, **earlier** than the admission date of $\text{Jan } 15$. This violates the rule that $\text{Discharge Date} \geq \text{Admission Date}$.

*Structural inconsistency*

Suppose that under the weight column, patient $\text{P001}$ has a value of $70$ and $\text{P002}$ has a value of $155$.

- Was it 70 kilograms and 155 kilograms?
- Or 70 kg and 155 pounds?

**Without consistent** units, **analysis** becomes **unreliable**.

*Timeliness*

Suppose a patient's diagnosis changes (**Table 13**). The latency is **3 months**. A prediction made in February would incorrectly assume the patient was non-diabetic.

**Table 13: Illustration of delayed data updating.**

| Event | Date |
|---|---|
| Patient develops diabetes | January |
| Database updated | April |

*Stale data*

Suppose **last cholesterol measurement** was made in **January,** and **prediction** was made in **October**. Feature age is given by $\Delta t = 10 - 1 = 9$ months. *If clinical guidelines require measurements newer than six months, the feature is stale*.

*Temporal leakage*

Suppose we predict hospital readmission **on discharge day**. The training data accidentally included: *Whether the patient attended a follow-up appointment two weeks later*. This information **did not exist at prediction time**. The model achieves 99% accuracy during evaluation but **fails in deployment** because it learned from future information.

Table 14: Age-group distribution in the real population.

| Age Group | Real Population |
|---|---|
| 18–40 | 40% |
| 41–65 | 35% |
| 65+ | 25% |

*Representativeness*

Suppose the target population is shown in Table 14 and the collected dataset in Table 15.

Table 15: Age-group distribution represented in the dataset.

| Age Group | Dataset |
|---|---|
| 18–40 | 5% |
| 41–65 | 15% |
| 65+ | 80% |

Although there are **one million patients**, the dataset is heavily skewed toward older adults. Adding another million elderly patients **does not improve representativeness**.

*Intersectional underrepresentation*

Consider the data shown in Table 16.

Table 16: Cross-tabulation of age group by gender.

| Group | Young | Elderly |
|---|---|---|
| Men | 4500 | 500 |
| Women | 500 | 4500 |

In terms of gender the dataset looks balanced i.e., equal representation of male and female. However, **at the next level of resolution**, young women and elderly men are **severely underrepresented**. Overall gender balance hides an important representativeness problem.

*Random measurement error*

Suppose a blood pressure monitor fluctuates slightly. True values are 140, 140, 140, and 140. Recorded values are 138, 141, 142, and 139. **The errors vary randomly around zero** i.e., -2, +1, +2, -1. The **average remains approximately correct**.

*Systematic measurement error*

Suppose the monitor was never calibrated. The true values are 140, 130, and 120 mmHg and the recorded values are 150, 140, and 130 mmHg respectively. Every reading is 10 mmHg too high. **The measurements are consistently biased**. **No amount of averaging removes this error**.

*Differential measurement error*

Suppose a wearable heart-rate sensor performs differently depending on skin tone. For the light skin group, average error is $\pm 2$ bpm and for the dark skin group, average error is $\pm 10$ bpm. The model trained on these measurements **may perform worse for darker-skinned patients** because one subgroup is measured much less accurately.

*Construct validity*

Suppose a hospital wants to predict **which patients need the most medical care**. Instead of measuring **health need**, it uses **healthcare spending** (Table 17).

**Table 17: Illustration of proxy measurement in healthcare resource allocation.**

| Patient | Health Need | Healthcare Spending |
|---|---|---|
| A | High | $30,000 |
| B | High | $4,000 |

Patient B has the **same medical need** but **received less care** due to barriers in access. A model trained on spending will **incorrectly conclude** that Patient B has lower need.

The algorithm works exactly as designed. It is simply optimizing the wrong target. This illustrates a failure of **construct validity** i.e., the **chosen proxy** (spending) **does not faithfully represent** the intended concept (health need).

*Key lesson*

These examples show that each dimension of data quality captures a **distinct property**:

- **Completeness:** Is the required data present?
- **Accuracy:** Are the recorded values correct?
- **Consistency:** Do the values agree with each other and established rules?
- **Timeliness:** Is the data available when it is needed?
- **Representativeness:** Does the dataset reflect the population where the model will be deployed?
- **Measurement Error:** Does the observed variable faithfully measure the underlying construct?

# Data Profiling and Curation

Data curation is the systematic process of **acquiring**, **organizing**, **validating**, **documenting**, **preserving**, and **maintaining data** to ensure that they **remain accurate**, **accessible**, and **suitable** for their intended purposes throughout their lifecycle.

Although data cleaning is an important component of curation, the **two concepts are not synonymous**. Data cleaning focuses primarily on identifying and correcting errors, inconsistencies, and missing values. Data curation **encompasses a much broader set of activities** that support the **long-term quality**, **usability**, and **reproducibility of data**. A robust curation practice also includes:

- **Scope and Governance:** Determining exactly which data belongs in the collection, establishing strict definitions, managing access controls, and defining the legal basis for use.
- **Integrity and Lineage:** Resolving duplicates, documenting data provenance, recording explicit exclusions, and preserving reproducibility.
- **Lifecycle Management:** Maintaining rigorous version control and systematically retiring obsolete data.

### 2.4.1 What data curation entails

These activities include:

- determining which data should be included in the collection;
- defining variables, coding schemes, and metadata standards;
- identifying and resolving duplicate records;
- documenting data provenance and collection history;
- maintaining version histories;
- controlling access and permissions;
- recording exclusions and transformation decisions;
- preserving reproducibility through comprehensive documentation; and
- archiving or retiring obsolete datasets while maintaining historical records.

Well-curated data are easier to understand, reuse, reproduce, and maintain over time.

### 2.4.2 A principled data-curation workflow

Data curation should follow a **structured** and **reproducible workflow** rather than an ad hoc sequence of corrections. Although specific procedures vary across organizations, a **general workflow** is depicted in **Figure 15**.




**Figure 15: Data Governance and Quality Workflow.** The figure presents an eight-stage workflow for transforming raw data into a reliable operational asset: inventorying data resources, profiling their characteristics, validating quality and conformance, cleaning identified defects, reconciling inconsistencies across sources, documenting definitions and provenance, versioning approved artifacts, and granting formal release approval. Together, these stages support reproducibility, accountability, traceability, and dependable downstream analytics and AI development.

### 2.4.3 Inventory

Before modifying a single cell, create a comprehensive manifest detailing the data's origins and constraints i.e., **Provenance and Ownership**, **Scope and Structure**, and **Governance**. Typical information recorded includes:

- data source;
- data owner or custodian;
- purpose of data collection;
- temporal coverage;
- file format and storage location;
- unit of analysis;
- approximate dataset size;
- sensitive or protected fields;
- access permissions and restrictions;
- legal or regulatory basis for collection; and
- expected update schedule.

### 2.4.4 Data Profiling

Data profiling examines the **statistical** and **structural characteristics** of a dataset to **identify potential quality issues** before cleaning begins. It inspects **basic metrics**, **quality indicators**, **distributions**, and **composition**. Typical profiling statistics include:

- numbers of rows and columns;
- data types;
- counts of unique values;
- proportions of missing data;
- variable distributions;
- minimum and maximum values;
- class or label balance;
- duplicate records;
- temporal coverage;
- and demographic or subgroup composition.

Profiling helps identify **anomalies**, **inconsistencies**, and **unexpected patterns** that warrant further investigation.

### 2.4.5 Validate

Validation determines whether the dataset **satisfies predefined structural**, **logical**, and **domain-specific** requirements. It performs **structural tests**, **relational tests**, and **logical tests**. Typical validation checks include:

- schema conformity;
- acceptable value ranges;
- uniqueness of primary keys;
- referential integrity between related tables;
- logical consistency among variables;
- correct chronological ordering of timestamps;
- and agreement across multiple data sources.

Many validation rules **can be automated**, enabling **systematic detection** of **quality problems** before analysis.

### 2.4.6 Record every change (immutable audit trails)

Data cleaning **must never overwrite** raw data **without absolute traceability**. The original raw data must be **preserved in an immutable state**. Every modification should be transparent, reproducible, and attributable. For each change, the following information should be recorded:

- the original value;
- the corrected value;
- the rule or procedure used to make the correction;
- the date and time of the modification;
- the software or tool version used;
- the individual responsible for the change;
- and the justification for the modification.

Maintaining **comprehensive change logs** ensures **reproducibility**, **facilitates quality assurance**, and allows future users to **understand exactly** how the curated dataset was produced.

# Data Cleaning

Data cleaning is the process of **identifying**, **investigating**, and **correcting errors**, **inconsistencies**, and **quality issues** that may **compromise the reliability** of a dataset. It is a critical stage of the data preparation pipeline because the **validity of statistical analyses** and **machine learning models** depends heavily on the **quality** of the **underlying data**. Effective data cleaning extends beyond simply removing incorrect values; it requires **understanding the causes** of missing data, **distinguishing** genuine observations from errors, **resolving** duplicate records, **standardizing** labels and measurement units, and **addressing** class imbalance while **preserving information that is meaningful** for the intended analysis. We examine the major challenges encountered during data cleaning and present principled methods for improving data quality **without introducing bias** or **information leakage**.

## 2.5.1 Missing values

Missing values occur when **one or more observations** are **unavailable**. Their presence is common in real-world datasets and may arise for many different reasons. Understanding **why data are missing** is often **as important as identifying where they** are missing, since the **underlying mechanism** influences the appropriate **analytical treatment**. Common causes of missing values include:

- a measurement was never taken;
- a respondent declined to answer;
- sensor or equipment failure;
- a clinical test was deemed unnecessary;
- a system failed to record the information;
- unsuccessful record linkage;
- deliberate removal for privacy or confidentiality;
- and fields that are not applicable to a particular observation.

**Not all** missing values **indicate poor data quality**; in many situations, they are **expected** and **informative**.

### Missing-data indicators

Missingness can be represented by an indicator matrix.

$$R_{ij} = \begin{cases} 1, & \text{if } X_{ij} \text{ is observed} \\ 0, & \text{if } X_{ij} \text{ is missing} \end{cases}$$

The pattern of the **indicator matrix** $R$ may itself **contain useful information**. For example, patients who undergo more laboratory tests are often systematically different from those who receive fewer tests.

### Classical missingness mechanisms

Statistical analyses commonly distinguish three mechanisms describing how missing values arise.

*Missing Completely at Random (MCAR)*

$$P(R|X_{\text{observed}}, X_{\text{missing}}) = P(R)$$

Under MCAR, the **probability of missingness** is **unrelated** to either **observed** or **unobserved data**.

*Missing at Random (MAR)*

$$P(R|X_{\text{observed}}, X_{\text{missing}}) = P(R|X_{\text{observed}})$$

Here, missingness **may depend** on **observed variables** but not on the missing values themselves after conditioning on the observed data.

*Missing Not at Random (MNAR)*

Under MNAR, the **probability of missingness** continues to **depend on** the **unobserved values** even after accounting for observed information. These mechanisms **describe assumptions** about the **data-generating process** rather than properties that can usually be verified directly from the dataset.

*Example # 45: Missingness occurs purely by chance*

**Lost questionnaires:** A researcher collects 500 survey forms. During transport, 30 questionnaires are accidentally lost because a box falls off a truck. The missing surveys are random and have nothing to do with respondents' ages, incomes, or answers.

**Random equipment failure:** A laboratory instrument fails randomly for 5% of blood samples because of an electrical glitch. The failure is unrelated to patients' health, age, or blood chemistry.

*Example # 46: Missingness can be predicted using information that is already observed*

**Income missing by age:** A survey records age and income. Younger participants are less likely to report their income than older participants.

- *Observed variable:* Age
- *Missing variable:* Income

Once age is known whether income is missing does not depend on the actual income value.

**Follow-up visits:** In a medical study, older patients are more likely to miss their follow-up appointment. Cholesterol measurements are therefore missing more often for older participants.

- *Observed variable:* Age
- *Missing variable:* Follow-up cholesterol

The missingness is explained by age, an observed variable.

*Example # 47: The missing value influences whether it is missing*

**High-income respondents:** People with very high incomes are less likely to report their income because of privacy concerns.

Even after adjusting for age, education, and occupation, individuals with higher incomes remain more likely to leave the income question blank.

- *Missing variable:* Income
- *Reason for missingness:* The income value itself

**Depression severity:** Patients experiencing severe depression are less likely to complete a mental health questionnaire because of their symptoms.

Even after accounting for age, sex, and previous diagnoses, those with the most severe depression are still the most likely to have missing responses.

- *Missing variable:* Depression score
- *Reason for missingness:* The unobserved severity itself

### Approaches to handling missing values

The appropriate strategy depends on the amount, pattern, and mechanism of missingness, as well as the intended analysis. Common approaches include:

- retaining missing values when supported by the analysis method;
- introducing explicit missing-value indicators;
- simple imputation (e.g., mean or median);
- subgroup- or groupwise imputation;
- multiple imputation;
- model-based imputation;
- removing variables with excessive missingness;
- and collecting additional data whenever feasible.

No single approach is universally optimal.

### Simple imputation

For **numeric** features, impute with **mean**, if the feature is roughly symmetric and missingness is not tied to extremes, and with **median** in case of outliers or skewed distributions (common in income, lab values, session durations). For **categorical** features, impute with **mode** of the remaining feature entries.

*Example # 48: Missing device information*

Suppose the feature under study is `device_type` and it is missing for $2\%$. Mode imputation might fill in `Android` if it is the most common device type.

*Note*: Missingness itself can be informative. For instance, binary `is_missing` indicators are often useful features. Instructive examples include system logs where absence of an event itself is informative or optional fields in forms where skipping correlates with behavior or intent.

### Model-based imputation

Useful when **missingness is moderate**, the **feature is predictable** from other features, and **enough data** is available.

*Example # 49: Missing clinical marker*

If a patient's BMI is missing but height, weight, age, and gender are available, a ***k*-NN**[210] or **regression imputer**[211] can estimate BMI using the values of **similar patients** instead of just filling in the median. This **preserves relationships** between features and produces more **realistic**, **personalized imputations** for modeling.

*Caution*: If the missingness itself is informative (e.g., lactate ordered only for critically ill patients[212]), model-based imputation can **hide the clinical meaning of missingness**.

#### Limitations of simple imputation

Although simple imputation methods are **computationally convenient**, they may **introduce undesirable statistical effects**. For example, mean imputation can:

- reduce the observed variance;
- distort relationships among variables;
- create artificial clusters of identical values;
- and obscure meaningful differences between subgroups.

To avoid information leakage, imputation parameters **must be estimated** using **only the training data** and then **applied unchanged** to **validation** and **test datasets**.

### 2.5.2 Outliers

An outlier is an **observation** that **differs substantially** from the **expected pattern** of the data. Outliers may represent:

- genuine but rare observations;
- data-entry mistakes;

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[210] ***k*-NN (*k*-Nearest Neighbors) imputer:** A missing-value imputation method that estimates a value using the corresponding values from the $k$ most similar records based on other available features. Example: If a patient's BMI is missing, a 5-NN imputer finds the five patients with the most similar height, weight, age, and gender, then imputes BMI using the average (or weighted average) of their BMIs.

[211] **Regression imputer:** A missing-value imputation method that uses a regression model trained on complete cases to predict the missing value from other features. Example: Train a linear regression model to predict BMI from height, weight, age, and gender, then use the model's predicted BMI for patients with missing BMI.

[212] Lactate test is primarily ordered for patients suspected to be critically ill, so its absence often signals lower severity. It is not ordered for everyone because doing so would waste money, staff time, and patient resources. This selective ordering makes missingness informative. Hence, filling in missing values blindly can erase this clinical signal in models.

- incorrect measurement units;
- sensor malfunction;
- previously unseen populations;
- fraudulent activity;
- or changes in the underlying data distribution.

Consequently, outliers should be investigated rather than automatically removed.

### Statistical detection

One common approach standardizes observations using the $z$-score:

$$z_i = \frac{x_i - \bar{x}}{s}$$

where $\bar{x}$ is the sample mean and $s$ is the sample standard deviation. A more robust alternative uses the **interquartile range (IQR)**[213]:

$$\text{IQR} = Q_3 - Q_1$$

A frequently used screening rule identifies observations outside:

$$[Q_1 - 1.5\ \text{IQR}, Q_3 - 1.5\ \text{IQR}]$$

These methods identify observations for further investigation rather than providing automatic criteria for deletion.

### Multivariate outliers

An observation **may appear ordinary** when each variable is **considered individually** while remaining highly unusual when variables are examined jointly. Examples include:

- a young individual with an implausibly long employment history;
- a small financial transaction occurring in an impossible geographic sequence;
- combinations of clinical measurements that are physiologically incompatible.

[213] **Interquartile Range (IQR):** A measure of statistical dispersion defined as the difference between the third quartile ($Q_3$) and the first quartile ($Q_1$), representing the spread of the middle 50% of the data. Example: If $Q_1 = 22$ and $Q_3 = 30$, then $\text{IQR} = 30 - 22 = 8$. Values below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$ are typically considered outliers.

Detecting multivariate outliers typically **requires methods** that consider **relationships among variables** rather than individual distributions alone.

### Appropriate responses

Possible actions after identifying an outlier include:

- verifying the original data source;
- correcting documented errors;
- converting incorrect measurement units;
- retaining valid rare observations while using robust statistical methods;
- Winsorizing values when appropriately justified;
- creating separate categories for unusual observations;
- and excluding observations only according to predefined and documented criteria.

**Rare observations** are often the most important examples for machine learning systems. Data cleaning should **remove errors**, **not meaningful complexity**.

### Common strategies

- **Clipping/Winsorization:** It limits extreme values to a maximum/minimum threshold to reduce their impact. This keeps extremes from dominating without deleting them. For example, capping transaction amounts above $10,000.
- **Transformation (log, Box-Cox[214]):** Applies a mathematical transformation to reduce skew and compress outliers. For example, log-transforming income to handle very high earners.

  The **Box–Cox** transformation for a positive variable $x > 0$ is defined as follows:

$$x_{\text{mod}}(\lambda) = \begin{cases} \dfrac{x^{\lambda} - 1}{\lambda}; & \lambda \neq 0 \\ \ln(x) \quad ; & \lambda = 0 \end{cases}$$

[214] **Box–Cox** is a family of power transformations introduced by **George Box** and **David Cox** in 1964 to stabilize variance and make data more normally distributed. It finds an optimal exponent $\lambda$ to reduce skew and compress extreme values.

If $\lambda = 1$, the data is unchanged and when $\lambda = 0$ it becomes a log-transform. Different values of $\lambda$ induce different effects (**Table 18**).

**Table 18: Impact of $\lambda$ in Box-Cox Transformation.**

| $\lambda$ | Effect |
|---|---|
| **2** | stronger stretching of large values |
| **1** | approximately linear transformation |
| **0.5** | square-root type transformation |
| **0** | approaches log transformation |

For negative values, one can use the **Shifted Box-Cox** transformation i.e.,

$$x_{\text{mod}}(\lambda) = \begin{cases} \dfrac{(x+c)^{\lambda} - 1}{\lambda}; & \lambda \neq 0 \\ \ln(x+c) \quad ; & \lambda = 0 \end{cases}$$

- **Robust Loss Function:** Instead of modifying the data, these approaches adjust the loss function to reduce the influence of outliers.
  - **Huber Loss**[215] combines squared error for small residuals and absolute error for large residuals, making the model less sensitive to extreme values.

$$\mathcal{L}_{\delta}(e) = \begin{cases} \dfrac{1}{2}e^2 & ; \quad |e| \leq \delta \\ \delta\left(|e| - \dfrac{1}{2}\delta\right) & ; \quad |e| > \delta \end{cases}$$

  - **Quantile loss**[216] focuses on predicting a specific quantile of the target distribution, which naturally downweights the effect of extreme residuals. The **quantile loss** (also called **pinball loss**) for a given quantile $\tau \in (0,1)$ is defined as follows:

$$\mathcal{L}_{\tau}(y, \hat{y}) = \begin{cases} \tau(y - \hat{y}) & ; \quad y \geq \hat{y} \\ (1-\tau)\big((\hat{y} - y)\big) & ; \quad y < \hat{y} \end{cases}$$

[215] **Huber loss**, introduced by **Peter J. Huber** in **1964**, is a robust loss function that combines squared error for small residuals and absolute error for large residuals.

[216] **Quantile loss** was formalized by **Roger Koenker** and **Gilbert Bassett** in their seminal **1978** paper on regression quantiles, the quantile loss function (often called pinball loss) enabled robust modeling of conditional quantiles beyond the mean. It was further popularized by **Roger Koenker** and **Kevin F. Hallock**, in **2001**, as a loss function that focuses on estimating a specific quantile of the target distribution, reducing the impact of outliers.

where $y$ is the true value and $\hat{y}$ is the predictive value. It **penalizes underestimation and overestimation asymmetrically**, according to the chosen quantile ($\tau$).

*Pros and Cons*: They improve stability and generalization when data contains noise or rare extreme values. However, they can slow convergence, require careful tuning (e.g., threshold in Huber loss), and may underfit if outliers actually carry meaningful signal.

#### Important Warning

There are many *interesting* or *pathological* events of interest which often **lie in the tails** of data distribution. For example, fraud cases, rare equipment failures, or ICU deterioration[217] events are **rare compared to normal behavior**, so they sit in the **extreme (tail) ends of the distribution** rather than near the average.

### 2.5.3 Duplicate records

Duplicate records occur when multiple entries represent the same observation or entity. Duplicates commonly arise from:

- repeated data ingestion;
- repeated submission attempts;
- integration of multiple source systems;
- manual data entry;
- copied benchmark datasets;
- and errors in entity resolution.

#### Exact and near duplicates

**Exact duplicates** are identical across all relevant fields. **Near duplicates** differ only slightly because of **spelling variations**, **formatting differences**, **timestamps**, **abbreviations**, or **minor recording** inconsistencies. Near duplicates are generally more difficult to detect and often require approximate matching methods.

[217] An ICU deterioration event is a significant worsening of a critically ill patient's condition that requires urgent intervention. Examples include respiratory failure requiring intubation, sudden drop in blood pressure requiring vasopressors, cardiac arrest, severe sepsis progression, or transfer to a higher level of life support. In predictive modeling, these are often the adverse events the model tries to forecast early so clinicians can intervene sooner.

*Example # 50: Duplicates in medical record*

- **Exact duplicate:** The first two records (**Table 18**) are identical across all fields (Patient ID, name, age, gender, and diagnosis).
- **Near duplicate:** The last two records (**Table 18**) refer to the same patient but differ slightly in the name (Sara Ahmed vs. Sara A. Ahmed), making them harder to identify without **approximate matching techniques** such as **string similarity** or **record linkage**.

**Table 18: Illustration of duplicate and potentially inconsistent patient records.**

| Patient ID | Name | Age | Gender | Diagnosis |
|---|---|---|---|---|
| 101 | Ali Khan | 45 | Male | Diabetes |
| 101 | Ali Khan | 45 | Male | Diabetes |
| 102 | Sara Ahmed | 38 | Female | Hypertension |
| 103 | Sara A. Ahmed | 38 | Female | Hypertension |

## Why duplicates matter

Duplicate records can:

- **bias** estimated frequencies;
- give **disproportionate weight** to certain observations;
- **inflate** the apparent sample size;
- **contaminate** training and testing partitions;
- and produce **overly optimistic** evaluation results.

Removing duplicates is therefore an important step in ensuring **reliable statistical inference** and **model evaluation**.

## Entity resolution

Entity resolution seeks to determine **whether multiple records refer to the same underlying entity**. Information used for matching may include:

- institutional identifiers;
- names;
- dates of birth;

- addresses;
- telephone numbers;
- device identifiers;
- and behavioral or transaction patterns.

Because **matching is imperfect**, entity resolution may introduce both **false merges** (combining different entities) and **false splits** (failing to combine identical entities), particularly when records are incomplete or inconsistent.

### Inconsistent labels

Labels may become inconsistent for many reasons, including:

- disagreement among annotators;
- evolving labeling criteria;
- institutional differences;
- inherently ambiguous cases;
- changes in taxonomy[218];
- errors in automated labeling systems;
- and observations with multiple valid labels.

**Careful label management** is essential because supervised learning **algorithms rely on label quality**.

### Inter-annotator agreement

When multiple annotators assign categorical labels, agreement can be evaluated using measures such as:

- raw percentage agreement[219,220];

[218] **Taxonomy:** A structured system for **organizing and classifying data** into **predefined categories** or **labels**. Example: In a medical dataset, a diagnosis taxonomy may classify diseases into categories such as *Cardiovascular*, *Respiratory*, *Endocrine*, and *Neurological*. If the taxonomy changes (e.g., splitting *Diabetes* into Type 1 and Type 2), previously assigned labels may become inconsistent.

[219] **Raw agreement:** The proportion of instances on which annotators **assign the same label, without adjusting for agreement** that **could occur by chance**. Example: If two annotators agree on 90 out of 100 patient diagnoses, the raw agreement is 90%.

[220] J. W. Holley and J. P. Guilford, A note on the *G* index of agreement, Educational and Psychological Measurement, 24(4), 749-753,1964.

- Cohen's $\kappa$[221,222]: $\kappa = \frac{P_o - P_e}{1 - P_e}$ where $P_0$ is the observed proportion of agreement between the two raters and $P_e$ is the expected proportion of agreement by chance. $P_e = \sum_{i=1}^{k} P_{Ai} \times P_{Bi}$ where $k$ is the number of categories, $P_{Ai}$ is the proportion of cases assigned to category $i$ by Rater A, and $P_{Bi}$ is the proportion of cases assigned to category $i$ by Rater B.
- Fleiss' $\kappa$[223,224]: $\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$ where $\bar{P} = \frac{1}{N} \sum_{i=1}^{N} P_i$ is the average observed agreement among raters across all subjects and $\bar{P}_e$ is the expected agreement by chance, based on the overall category proportions. The observed agreement for subject $i$ is computed as follows: $P_i = \frac{1}{n(n-1)} \left( \sum_{j=1}^{k} n_{ij}^2 - n \right)$ where $n$ is the number of raters per subject and $n_{ij}$ is the number of raters who assign subject $i$ to category $j$. The expected chance agreement $\bar{P}_e = \sum_{j=1}^{k} p_j^2$ where $k$ is the number of categories and $p_j$ is the proportion of all ratings assigned to category $j$.
- adjudication outcomes[225,226];
- and class-specific disagreement rates[227,228].

**Agreement statistics** provide insight into **annotation reliability** but should always be **interpreted alongside** domain knowledge.

### Consensus does not always equal truth

[221] **Cohen's $\kappa$ (Cohen's kappa):** A statistical measure of **inter-annotator agreement** between two annotators that **adjusts for agreement expected by chance**. Example: Two radiologists independently classify 100 X-rays as normal or abnormal and achieve a Cohen's $\kappa$ of 0.82, indicating strong agreement beyond chance.

[222] Jacob Cohen, A coefficient of agreement for nominal scales, Educational and Psychological Measurement, 20(1), pp 37-46, 1960.

[223] **Fleiss' $\kappa$ (Fleiss' kappa):** A statistical measure of **inter-annotator agreement among three or more annotators** that **adjusts for chance agreement**. Example: Five clinicians independently label 200 skin lesion images as benign, malignant, or uncertain. A Fleiss' $\kappa$ of 0.76 indicates substantial agreement across the group.

[224] Joseph L. Fleiss, *Measuring nominal scale agreement among many raters*, Psychological Bulletin, 76(5), 378-382., 1971.

[225] **Adjudication outcomes:** The final **labels assigned** after **resolving disagreements** among annotators through **expert review** or **consensus**. Example: Two clinicians disagree on whether a chest X-ray shows pneumonia; after review by a senior radiologist, the final adjudicated label is Pneumonia, which becomes the adjudication outcome.

[226] G. Hripcsak and A. S. Rothschild, *Agreement, the F-measure, and reliability in information retrieval*, Journal of the American Medical Informatics Association, 12(3), 296-298, 2005.

[227] **Class-specific disagreement rates:** The **proportion of annotation disagreements** calculated **separately for each class (category)** to identify labels that are **more prone to inconsistency**. Example: Annotators disagree on 8% of Diabetes cases but 25% of Pre-diabetes cases, indicating a higher class-specific disagreement rate for Pre-diabetes.

[228] R. Artstein, and M. Poesio, *Inter-coder agreement for computational linguistics,* Computational Linguistics, 34(4), 555-596, 2008.

**Majority voting** provides a practical method for **producing consensus labels** but may **suppress legitimate expert disagreement** or **overlook genuinely ambiguous cases**. Annotation documentation should therefore preserve:

- original annotations;
- annotator expertise;
- consensus labels;
- adjudication decisions;
- uncertainty assessments;
- and acceptable alternative labels.

Maintaining this information supports transparency and future re-evaluation.

*Example # 51: Acceptable alternative labels*[229,230,231,232]

A skin lesion image is independently reviewed by three dermatologists (**Table 19**).

**Table 19: Illustration of inter-annotator disagreement in clinical image labeling.**

| Annotator | Label |
|---|---|
| Expert 1 | Benign nevus |
| Expert 2 | Dysplastic nevus |
| Expert 3 | Dysplastic nevus |

[229] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghgoo, Robyn Ball, Katie Shpanskaya, Jayne Seekins, David A. Mong, Safwan S. Halabi, Jesse K. Sandberg, Ricky Jones, David B. Larson, Curtis P. Langlotz, Bhavik N. Patel, Matthew P. Lungren, and Andrew Y. Ng, CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison, Proceedings of the AAAI conference on artificial intelligence, 2019.

[230] Jasper Linmans, Emiel Hoogeboom, Jeroen van der Laak, and Geert Litjens, The Latent Doctor Model for Modeling Inter-Observer Variability, IEEE journal of biomedical and health informatics, 2023.

[231] Lie Ju, Xin Wang, Lin Wang, Dwarikanath Mahapatra, Xin Zhao, Quan Zhou, Tongliang Liu, and Zongyuan Ge, Improving Medical Images Classification with Label Noise Using Dual-Uncertainty Estimation. IEEE Transactions on Medical Imaging, 41(6), 1533-1546, 2022.

[232] Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Re, *Hidden Stratification Causes Clinically Meaningful Failures in Machine Learning for Medical Imaging*, Proceedings of the ACM conference on health, inference, and learning, 2020.

After adjudication, the **consensus label** is **dysplastic nevus**. However, because the lesion exhibits features consistent with both categories and experts consider either diagnosis clinically reasonable, **Benign nevus** is retained as an **acceptable alternative label**.

- **Consensus label:** Dysplastic nevus[233]
- **Acceptable alternative label:** Benign nevus[234]

Recording the alternative label **preserves diagnostic uncertainty** and allows future studies, improved imaging, or **updated clinical guidelines** to re-evaluate the case without losing valuable expert judgment.

Note: Dermatopathologists may disagree because the **boundary between typical and atypical lesions** can be **subjective** and **influenced** by **experience**, **diagnostic criteria**, and **specimen characteristics**. Therefore, if one expert classifies the lesion as dysplastic nevus and another as benign nevus, the disagreement may reflect **legitimate diagnostic uncertainty**, not necessarily an error.

### Test-set label quality

Label quality is **especially critical** for **benchmark test datasets** because model rankings depend directly on these labels. Northcutt and colleagues famously demonstrated that widely used benchmark test sets (in vision, language, and audio) contained **pervasive label errors**[235]. When they corrected these labels, the **relative performance rankings** of state-of-the-art models **actually changed**, proving that **ground truth is often flawed**.

### Unit-aware schemas

Every numerical variable should include accompanying metadata describing:

- the physical quantity being measured;
- measurement units;
- valid ranges;
- measurement precision;

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[233] A **dysplastic nevus** is a type of melanocytic nevus with atypical clinical or histological features (e.g., irregular borders, color variation, architectural atypia).

[234] A **benign nevus** may show some atypical-looking features while still lacking sufficient evidence for dysplasia.

[235] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller, *Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks*, NeurIPS, 2021.

- conversion rules;
- and source-system conventions.

**Explicit documentation** greatly **reduces the likelihood** of unit-related errors.

#### Dimensional analysis

Mathematical operations involving physical quantities should **remain dimensionally consistent**. Whenever **variables are combined**, the resulting units should be **understood and documented**. For example,

$$x_{\text{kg}} = 0.45359237 x_{\text{lb}}$$

Unit conversions should **always be explicit** rather than inferred automatically, as **silent assumptions** frequently lead to **serious analytical errors**.

### 2.5.4 Class imbalance

In classification problems, **class prevalence** is defined as, $\pi_k = \frac{n_k}{n}$ where $n_k$ denotes the **number of observations in class** $k$ and $n$ is the **total number of observations**. Class imbalance occurs when one or more classes are substantially less frequent than others. Examples include:

- fraudulent transactions;
- rare diseases;
- equipment failures;
- safety incidents;
- and cybersecurity attacks.

#### Why accuracy can be misleading

**Overall accuracy** may provide an **overly optimistic assessment** of performance when **classes** are **highly imbalanced**. For example, if 99% of transactions are legitimate, a classifier that always predicts *legitimate* achieves an accuracy of 99% while failing to detect any fraudulent transactions. Consequently, accuracy alone is **often an inadequate evaluation metric** for **imbalanced datasets**.

#### Strategies for addressing class imbalance

Common approaches include:

- selecting evaluation metrics appropriate for imbalanced data;
- applying class-weighted learning;
- adjusting decision thresholds;
- over- or undersampling;
- collecting additional minority-class observations;
- using stratified data partitioning;
- reformulating the problem as anomaly detection;
- and employing cost-sensitive learning.

The most appropriate strategy depends on the application and the relative costs of different types of prediction errors.

### Risks of resampling

Oversampling[236] **before partitioning** the dataset may place **duplicated or synthetic observations** into both the **training** and **testing** sets, resulting in **overly optimistic estimates**[237] of model performance.

Resampling should therefore be performed **only within the training partition** or within **each training fold during cross-validation**[238].

*Example # 52: Risk of resampling in medical image analysis*

Suppose a medical dataset contains 1,000 patients: 950 Healthy and 50 Cancer.

*Incorrect approach (oversampling before partitioning)*

- Apply oversampling (e.g., duplicate or generate synthetic cancer cases) to balance the dataset.
- Split the balanced dataset into **80% training** and **20% testing** (Table 20).

[236] **Oversampling:** A **data balancing technique** that **increases** the number of **minority-class samples** by **duplicating existing observations** or **generating synthetic** ones to reduce class imbalance. Example: In a dataset with 950 healthy patients and 50 cancer patients, SMOTE generates synthetic cancer cases so that the minority class is better represented during model training.

[237] **Overly optimistic estimates (of model performance):** Performance metrics that are **artificially inflated** because information from the test data **unintentionally influences** model **training or evaluation**, leading to an **unrealistic assessment of generalization**. Example: If oversampling is performed before splitting the dataset, duplicated patient records may appear in both the training and test sets, resulting in an accuracy of 98% that is unlikely to be achieved on truly unseen data.

[238] **Cross-validation:** A model evaluation technique that **repeatedly partitions the dataset** into training and validation subsets to estimate **how well a model generalizes** to unseen data. Example: In 5-fold cross-validation, the dataset is divided into five equal parts; the model is trained on four folds and tested on the remaining fold, repeating the process five times so that each fold serves as the test set once.

Table 20: Class distribution after oversampling and subsequent train–test partitioning.

| Dataset | Healthy | Cancer |
|---|---|---|
| After oversampling | 950 | 950 |
| Training (80%) | 760 | 760 |
| Testing (20%) | 190 | 190 |

Because some duplicated or synthetic cancer samples are derived from the original minority-class observations, **very similar or identical samples may appear in both the training and testing sets**. The model effectively *sees* information from the test set during training, producing an **overly optimistic performance estimate** (e.g., **99% accuracy**).

*Incorrect approach (partition first, then oversample)*

- Split the original dataset into training and testing sets.
- Apply oversampling only to the training set.
- Leave the test set unchanged for evaluation (**Table 21**).

Table 21: Class distribution before and after training-set oversampling.

| Dataset | Healthy | Cancer |
|---|---|---|
| Original | 950 | 50 |
| Training (80%) | 760 | 40 |
| Testing (20%) | 190 | 10 |
| After oversampling (training only) | 760 | 760 |
| Testing (unchanged) | 190 | 10 |

The model is trained on a balanced training set but evaluated on **previously unseen, untouched test data**, providing a **more reliable estimate** of real-world performance.

*Key lesson*

Always **split the dataset before resampling**, and during **k-fold cross-validation**, perform oversampling **independently within each training fold**, never before creating the folds.

## Class prevalence and sampling design

A balanced training dataset **may improve** the learning process, but the **observed class proportions** in the training data may **differ substantially** from those encountered during deployment.

When this occurs, predicted probabilities **often require recalibration** to reflect the **true prevalence of outcomes** in the target population. Balancing the training data improves learning but does not alter the underlying frequencies of events in real-world applications.

*Example # 53: Recalibrating probabilities*

Suppose a hospital dataset contains 10,000 patients: 9,500 Healthy (95%) and 500 Cancer (5%). To improve learning, the training data are **oversampled** so that the model is trained on **9,500 Healthy** and **9,500 Cancer (50% prevalence)**. After training, the model predicts that a new patient has a **70% probability of cancer**.

However, the model was trained on an artificially balanced dataset, whereas the **real-world prevalence of cancer is only 5%**. As a result, the **70% predicted probability is likely too high** and does not accurately represent the patient's **true risk** in the target population.

A **probability recalibration** method (e.g., **Platt scaling**[239,240] or **isotonic regression**[241,242]) adjusts the prediction to account for the true disease

[239] **Platt scaling:** A probability calibration method that **fits a logistic regression model** to a **classifier's output scores**, converting them into calibrated probabilities. Example: A support vector machine predicts a patient has a cancer score of 2.8; after Platt scaling, this score is converted into a calibrated probability of 0.74 (74%), which better reflects the true likelihood of cancer.

[240] J. Platt, *Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods*, Advances in Large Margin Classifiers, pp. 61–74, MIT Press, 1999.

[241] **Isotonic regression:** A non-parametric **probability calibration** method that **fits a monotonically increasing function** to map model scores to calibrated probabilities **without assuming a specific functional form**. Example: A classifier assigns a patient a cancer probability of 0.85, but after isotonic regression calibration using validation data, the probability is adjusted to 0.63, better matching the observed frequency of cancer among similar patients.

[242] B. Zadrozny and C. Elkan, *Transforming Classifier Scores into Accurate Multiclass Probability Estimates*, Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp 694-699, 2002.

prevalence. After recalibration, the same patient might receive a probability of **22%** instead of **70%**, providing a more realistic estimate of cancer risk.

### Platt scaling

**Platt scaling** is performed by training a **logistic regression model** on a classifier's output scores (not the original features) to convert those scores into calibrated probabilities.

*Procedure*

- **Train the original classifier.**
  - For example, train a Support Vector Machine (SVM) to classify patients as *Cancer* or *Healthy*.
- **Obtain prediction scores on a validation dataset.**
  - Instead of class labels, record the classifier's raw decision scores (e.g., SVM decision values).
- **Fit a logistic regression model.**
  - Use the raw scores as the predictor and the true class labels as the response.
  - The calibration model is:

$$P(y = 1|s) = \frac{1}{1 + \exp(As + B)}$$

  - where $s$ is the classifier score, $A$ and $B$ are parameters estimated on the validation data.
- **Apply the calibration model to new predictions.**
  - Every future classifier score is passed through the logistic function to produce a calibrated probability.

*Example # 54: Recalibrating probabilities with Platt scaling*

Suppose an SVM[243,244] produces the scores tabulated in **Table 22**. Platt scaling **learns the relationship** between these **scores** and the **true outcome**

[243] **Support Vector Machine (SVM):** A supervised machine learning algorithm that **classifies data by finding the optimal decision boundary (hyperplane) that maximizes the margin between classes**. Example: An SVM is trained to classify skin lesions as benign or malignant using features extracted from dermoscopic images. For a new lesion, the SVM produces a decision score (e.g., 1.6), which predicts the malignant class; this score can be converted into a calibrated probability using Platt scaling.

[244] C. Cortes and V. Vapnik, *Support-Vector Networks*, Machine Learning, 20(3), 273-297, 1995.

**frequencies**, producing probabilities that better reflect the actual likelihood of the event.

**Table 22: Illustrative SVM decision scores and true diagnostic labels.**

| Patient | SVM score | True label |
|---|---|---|
| A | 2.5 | Cancer |
| B | 1.8 | Cancer |
| C | 0.4 | Healthy |
| D | -1.2 | Healthy |

A logistic regression is fitted using the SVM scores and true labels. For a **new patient**: SVM score $= 1.6$ and raw output $= 1.6$ (not a probability). After applying the fitted Platt scaling model:

$$P(\text{cancer}) = \frac{1}{1 + \exp(A \times 1.6 + B)}$$

Suppose the estimated parameters are $A = -1.3$ and $B = 0.4$. Then

$$P(\text{cancer}) = \frac{1}{1 + \exp(-1.3 \times 1.6 + 0.4)} \approx 0.84$$

The **calibrated prediction** becomes: **Probability of cancer** $= 84\%$ instead of simply reporting the raw score of $1.6$.

## SMOTE and SMOTENC in Practice

**SMOTE**[245,246] (Synthetic Minority Over-sampling Technique) addresses imbalance by creating synthetic minority-class observations through interpolation between neighboring minority samples in continuous feature

[245] **SMOTE** (Synthetic Minority Over-sampling Technique) was introduced in **2002** by **Chawla et al.** as a method to address class imbalance in machine learning. It generates synthetic samples for the minority class by interpolating between existing examples. SMOTE quickly gained popularity and was extended with variations like **SMOTENC** (for categorical data) in 2003. Over time, many adaptations, such as **Borderline-SMOTE** and **ADASYN**, were developed to improve its effectiveness. Today, SMOTE remains a widely used technique in imbalanced data handling across various fields.

[246] **ADASYN** (Adaptive Synthetic Sampling) was introduced in **2008** by **He et al.** as an improvement over SMOTE. It adaptively generates more synthetic samples for minority class instances that are harder to classify, based on their proximity to majority class samples. ADASYN aims to focus on difficult-to-learn examples, improving model performance on imbalanced datasets.

space. Rather than simply duplicating rare cases, it expands the minority region more smoothly, often helping the classifier learn a broader and less fragmented boundary.

**SMOTENC**[247,248] extends this idea to datasets containing both numerical and categorical features. It preserves the integrity of categorical variables while synthesizing numerical components appropriately, making it suitable for tabular pipelines in which encoded or raw categorical predictors are present. In practice, the choice between SMOTE and SMOTENC is determined by feature composition:

- **SMOTE** is appropriate when all resampled features are continuous.
- **SMOTENC** is required when categorical and numerical variables coexist.

[247] **SMOTENC** (Synthetic Minority Over-sampling Technique for Nominal and Continuous) was introduced in **2003** by **Chawla et al.** as an extension of SMOTE to handle datasets with both **categorical (nominal)** and **continuous** features. It modifies the original SMOTE algorithm by treating categorical features differently, ensuring synthetic samples respect the distinct values of categorical variables. SMOTENC has since become a standard tool for imbalanced data in tasks involving mixed data types. Variants of SMOTENC, such as **Borderline-SMOTENC**, have also been developed to enhance performance in specific scenarios.

[248] **Borderline-SMOTENC** was introduced in **2005** by **Han et al.** as an extension of SMOTENC, designed to focus oversampling efforts on minority class instances near the decision boundary. It generates synthetic samples for these boundary instances to improve model performance in imbalanced datasets. This method aims to enhance the model's ability to correctly classify difficult-to-learn boundary samples.

# Data Leakage

Data leakage is one of the **most common** and **damaging sources of error** in predictive modeling. It occurs when **information unavailable** at the **intended prediction time** is **inadvertently used** during **model development**, leading to **overly optimistic estimates** of performance that **rarely generalize to real-world** deployment. Leakage can arise in many forms, including target leakage, train–test contamination, temporal leakage, and group leakage. Because leaked information **often produces models** that **appear highly accurate** during evaluation but **fail in practice**, preventing leakage **requires careful attention** to data partitioning, preprocessing, feature engineering, and experimental design. We examine the principal forms of data leakage, explains why they occur, and presents practical strategies for detecting and preventing them throughout the machine learning pipeline.

## 2.6.1 What is data leakage?

Data leakage occurs when **information** that would **not legitimately be available** at the intended prediction or evaluation time is **inadvertently used** during model development.

Leakage typically produces **overly optimistic estimates** of model performance because the model **exploits information that would not exist** when making predictions on new data. As a result, evaluation results may **substantially overestimate** real-world performance. A practical way to identify potential leakage is to ask:

*Could this feature, preprocessing statistic, or related record genuinely be available in the same form at the time a prediction is made for a new observation?*

If the answer is no, the information should not be used during model training or evaluation. Kaufman and colleagues[249] identified data leakage

[249] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, *Leakage in Data Mining: Formulation, Detection, and Avoidance*. ACM Transactions on Knowledge Discovery from Data, 6(4), 15, 2012.

as a **major yet often overlooked threat** to the validity of data mining and machine learning systems.

### 2.6.2 Target leakage

Target leakage occurs when **one or more input features** directly or indirectly **reveal the outcome** that the **model is intended to predict**. For example, suppose the objective is to predict hospital mortality at the time of admission. Features such as the following would constitute target leakage because they **become available only** after the prediction point:

- discharge disposition;
- postmortem procedures;
- final diagnoses established after death;
- treatment decisions made immediately before the outcome;
- and variables calculated using the outcome itself.

Formally, if a feature $X_j$ is generated using information from the target variable $Y$ after the prediction time, $Y \rightarrow X_j$, then using $X_j$ to predict $Y$ **violates the intended temporal** or **causal ordering** and introduces leakage.

#### Subtle target leakage

Target leakage is **not always obvious**. A feature **may never explicitly contain** the outcome label **yet still function as an almost perfect proxy** for it. Examples include:

- invoice status recorded after loan default;
- customer cancellation codes assigned after churn;
- medical interventions initiated only after disease recognition;
- clinical notes or reports written after the outcome occurred.

**Unexpectedly high** predictive performance should therefore **prompt careful investigation** rather than immediate confidence in the model.

*Example # 54: Sepsis prediction*

A hospital develops a model to predict whether a patient will develop **sepsis** within 24 hours. The dataset includes a feature named **Broad-spectrum antibiotics prescribed (Yes/No)**.

Although this feature does **not explicitly contain the sepsis diagnosis**, antibiotics are typically prescribed **only after clinicians suspect or recognize sepsis**. As a result, the model learns to rely on this feature as an almost perfect proxy for the outcome rather than learning genuine early risk factors.

The model achieves an AUC of 0.99, but this **unusually high performance** is misleading because the feature contains information that would **not be available at the time the prediction is intended to be made**.

### 2.6.3 Train-Test contamination

Train-test contamination occurs when **information from the evaluation dataset** influences **model development**, **parameter estimation**, or **model selection**. Common sources include:

- identical records appearing in both training and testing sets;
- near-duplicate images distributed across partitions;
- preprocessing fitted using the entire dataset;
- feature selection performed using test labels;
- repeated model tuning against the test set;
- pretraining on benchmark test examples;
- and synthetic observations generated from test records.

**Proper separation** between training and evaluation data is **essential for** obtaining **unbiased estimates** of model performance.

#### Preprocessing contamination

Data preprocessing must be learned **exclusively from the training data**. An incorrect approach computes the mean using all observations:

$$\mu = \frac{1}{n_{\text{all}}} \sum_{\text{all data}} x_i$$

The correct procedure **estimates** preprocessing parameters **only from the training set**:

$$\mu_{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{i \in \text{train}} x_i$$

and applies the resulting transformation to every partition:

$$x_i' = \frac{x_i - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

The same principle applies to all preprocessing operations.

### 2.6.4 Cross-validation

Within cross-validation, **every data-dependent transformation** must be fitted **independently inside** each **training fold**. This includes:

- missing-value imputation;
- feature scaling;
- categorical encoding;
- feature selection;
- oversampling or undersampling;
- dimensionality reduction;
- and threshold optimization.

Applying these procedures before creating cross-validation folds introduces information leakage and biases performance estimates.

Cross-validation (CV) **reduces instability** by evaluating across **multiple splits** and **averaging**. This gives:

- More **reliable model comparisons** than a single split.
- A sense of **variance across folds** (important for risk-aware decisions).

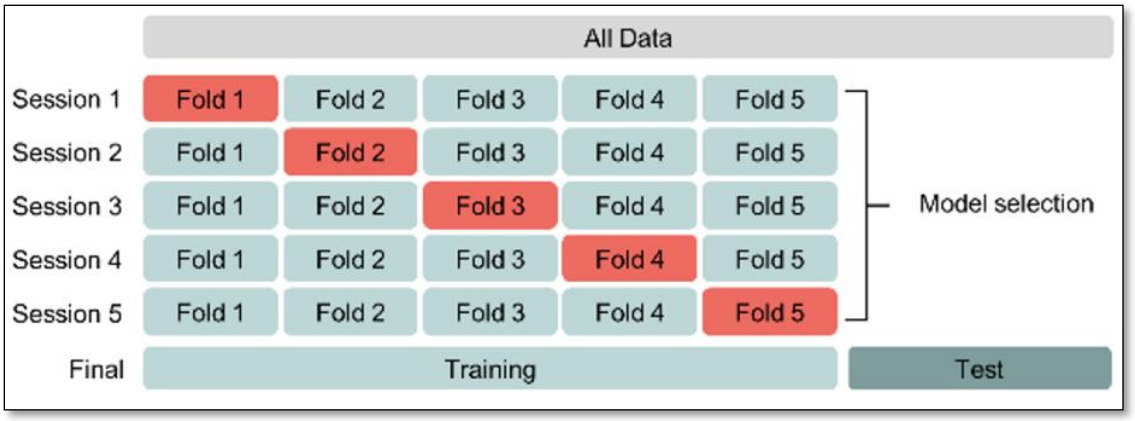



**Figure 16: Five-fold cross-validation with a hold-out test data set[250].**

[250] Tyler J. Bradshaw, Ronald Boellaard, Joyita Dutta, Abhinav K. Jha, Paul Jacobs, Quanzheng Li, Chi Liu, Arkadiusz Sitek, Babak Saboury, Peter J.H. Scott, Piotr J. Slomka, John J. Sunderland, Richard L. Wahl, Fereshteh Yousefirizi, Sven Zuehlsdorff, Arman Rahmim and Irène Buvat, *Nuclear Medicine and Artificial Intelligence: Best Practices for Algorithm Development*, Journal of Nuclear Medicine, 63, No. 4, pp 500-510, 2022.

### $k$-Fold Cross-validation

In $k$-fold CV (Figure 16), we partition the data into $k$ folds: **Train** on $k-1$ folds and **Validated** on the remaining fold. **Repeat** $k$ times, each fold serving as validation once. Average the metrics across $k$ folds (for mean and standard deviation of values).

- **Standard $k$-Fold CV:** Used when samples are approximately **independent and identically distributed** (*iid*) i.e., observations are independent of one another and drawn from the same underlying distribution. This means there is **no temporal order**, **grouping structure**, or **dependency** (e.g., repeated measures, users, time series). When this assumption holds, K-Fold provides a reliable estimate of generalization performance.

  KFold splits data into K sequential folds; each fold is used once for validation.

  RepeatedKFold repeats K-Fold multiple times with different random splits to reduce variance in performance estimates

- **Stratified $k$-Fold CV:** Designed specifically for **classification problems**, especially when classes are **imbalanced**. Instead of splitting the data purely at random, it ensures that **each fold preserves approximately the same class distribution as the full dataset**.

  This is **particularly important** when the **positive class is rare** (e.g., fraud detection, churn prediction, medical diagnosis). If standard $k$-Fold is used on highly imbalanced data, **some folds may end up with very few**, or even zero, positive examples. In such cases, **performance metrics** like precision, recall, and $F_1$-score become **unstable or meaningless**, because the model is not properly evaluated on the minority class. Stratified $k$-Fold produces **more reliable and representative performance estimates**, especially for metrics sensitive to class imbalance.

  StratifiedKFold is a $k$-fold variant that preserves class proportions in each fold (for classification).

  StratifiedShuffleSplit generates random train/test splits repeated multiple times while preserving class proportions.

- **Grouped CV:** Is used when **observations are not independent**, but instead **belong to identifiable groups** such as patients, users, devices, sessions, or households. In these cases, multiple rows may come from the same underlying entity, creating **strong within-group similarity**.

  If standard $k$-Fold is used, samples from the same group can appear in both the training and validation sets. This creates **data leakage at the group level**. The model may implicitly recognize characteristics of a specific patient or device rather than learning patterns that generalize to new, unseen groups. As a result, validation performance can appear optimistically high.

  Grouped CV prevents this by ensuring that **entire groups are assigned exclusively to either the training set or the validation set in each fold**. This forces the model to generalize across groups rather than memorize group-specific signals, leading to a more realistic estimate of real-world performance when predicting for new entities.

  `GroupKFold` prevents group overlap across folds.
  `StratifiedGroupKFold` tries to preserve class proportions while keeping groups separated.

- **Time-series CV:** Designed for datasets where **observations are temporally ordered**, such as stock prices, sensor readings, or customer behavior over time. Standard random splitting would **violate the chronological order**, potentially allowing the model to *see the future*, which creates data leakage and unrealistic performance estimates.

  `TimeSeriesSplit` splits data into sequential training and test sets, ensuring that the training set always comes before the validation set in time.

Few other modes of splitting include:
`LeaveOneOut` (LOO) uses all samples except one for training and tests on the **single left-out sample**; repeated for every observation.

`ShuffleSplit` randomly splits data into train/test sets multiple times, **without enforcing fold structure**.

### 2.6.5 Temporal leakage

Temporal leakage occurs when **information from the future** is used to **predict an earlier outcome**. If predictions are made at time $t$, every feature must satisfy $t_{\text{availability},j} \leq t$. The **relevant timestamp** is the moment at which the **information became available**, not necessarily when the underlying event occurred. Examples of temporal leakage include:

- using laboratory results finalized after the prediction time;
- randomly mixing future and past observations during dataset partitioning;
- incorporating revised economic statistics unavailable at the forecasting date;
- using maintenance records entered after equipment failure;
- and computing summary statistics using future observations.

These practices allow the model to exploit information that would not exist in operational settings. When models are intended for future deployment, evaluation should preserve chronological order (Figure 17). **Temporal partitioning** provides a **more realistic estimate** of future predictive performance than **random data splitting**.

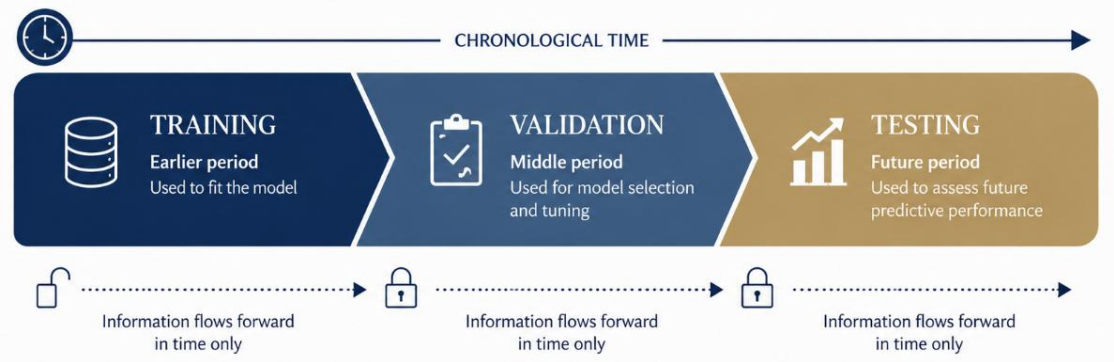



**Figure 17: Temporal evaluation design for future deployment.**

### 2.6.6 Group leakage

Group leakage occurs when **related observations** from the **same entity appear in multiple data partitions**. Groups may correspond to:

- patients;
- households;
- students;
- customers;
- devices;

- hospitals;
- geographic regions;
- videos;
- document sources;
- or repeated measurements from the same experimental subject.

Because **observations within a group** are often **highly correlated**, splitting them across training and testing datasets can **substantially inflate** measured performance.

Suppose several medical images from the same patient are randomly assigned to different data partitions. **Rather than learning generalizable** clinical patterns, the model **may simply recognize patient-specific** characteristics. To prevent this problem, the partitioning should ensure:

$$G_{\text{train}} \cap G_{\text{test}} = \phi$$

meaning that no entity appears in both training and testing datasets.

*Example # 55: Customer churn prediction*

A telecommunications company develops a machine learning model to predict whether a **customer will cancel their subscription (churn)** [251]. The dataset contains multiple records per customer because each customer has monthly usage, billing, and support interactions (**Table 23**).

**Table 23: Illustration of entity-level data leakage across training and testing partitions.**

| Customer ID | Record | Dataset partition |
|---|---|---|
| C001 | January usage data | Training |
| C001 | February usage data | Testing |
| C001 | March usage data | Training |
| C002 | January usage data | Training |
| C003 | January usage data | Testing |

[251] **Churn:** The loss or termination of a customer's relationship with a company, service, or product over a defined period, often used as a target variable in predictive modeling. Example: A telecom customer who cancels their mobile subscription is labeled as $\text{churn} = 1$, while a customer who continues using the service is labeled $\text{churn} = 0$.

Because **Customer C001 appears in both training and testing sets**, the model can **learn customer-specific patterns** (e.g., typical usage level, contract type, account history) that are unique to that individual **rather than learning** general churn patterns.

The model achieves 95% accuracy, but the **result is misleading** because the **test set contains information** from customers **already seen during training**.

The correct approach should follow a customer-level splitting. All records from the same customer should remain together (Table 24).

**Table 24: Entity-level partitioning of longitudinal customer records.**

| Customer ID | Records | Dataset partition |
|---|---|---|
| C001 | Jan, Feb, Mar usage data | Training |
| C002 | Jan usage data | Training |
| C003 | Jan usage data | Testing |

The model is then evaluated on **entirely new customers**, providing a more realistic estimate of how it will perform after deployment.

Hence, when multiple observations **belong to the same entity** (customer, patient, device, household, etc.), splitting should occur at the **group level**, not at the individual record level.

### 2.6.7 Site leakage

When deployment is intended for **previously unseen institutions**, random patient-level partitioning within the **same hospitals does not adequately evaluate generalization**. Instead, an **entire hospital or site should be held out** during evaluation so that testing reflects performance on **genuinely new clinical environments**.

### 2.6.8 Data leakage prevention checklist

Preventing leakage requires **careful planning** throughout the model development process. Before fitting a predictive model, verify that you have:

- clearly defined the prediction time;
- constructed the complete data timeline;
- identified all grouping variables and dependent entities;
- partitioned the data before fitting any preprocessing procedures;
- searched for duplicate and near-duplicate records;
- confirmed that target variables and derived fields are excluded from predictors;
- fitted all preprocessing operations using only the training data;
- performed resampling only within training partitions or training folds;
- preserved an entirely untouched test dataset for final evaluation;
- investigated variables with unexpectedly high predictive power;
- and documented precisely when each feature becomes available.

When deployment involves future time periods, new institutions, or previously unseen populations, the **evaluation design should mirror those deployment conditions** to provide an **unbiased assessment of real-world performance**.

# Data Documentation

A dataset **without adequate documentation** may be **impossible to evaluate**, **interpret**, or **use responsibly**. Documentation provides the context required to understand not only **what data are available**, but also **how they were created**, **what assumptions** shaped their construction, and **what limitations affect** their use. Comprehensive documentation enables users to determine:

- what information the dataset contains;
- why the dataset was created;
- how observations or participants were selected;
- how labels or annotations were produced;
- what preprocessing and transformations were applied;
- which applications are appropriate;
- which applications should be avoided;
- who is responsible for maintaining the dataset;
- and what ethical, legal, or governance restrictions apply.

Documentation **does not eliminate bias** or **guarantee responsible use**. However, it makes **important assumptions**, **design choices**, and **limitations transparent**, allowing users to make more informed decisions.

## 2.7.1 Datasheets for datasets

To improve **transparency** and **communication** between dataset **creators** and **users**, Gebru and colleagues proposed **datasheets for datasets**, inspired by documentation practices used for manufactured products[252].

A datasheet encourages dataset creators to **reflect systematically** on the motivation, construction, limitations, and appropriate uses of their datasets. It also **provides users with the information** needed to determine whether a dataset is suitable for a particular purpose. A comprehensive datasheet should address the following areas:

### Motivation

[252] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford, *Datasheets for Datasets*, arXiv, 2018.

Documentation should explain why the dataset was created. Important questions include:

- What problem or research need motivated the dataset?
- Who funded or supported its creation?
- What task, application, or knowledge gap does it address?
- Who are the intended users?

### Composition

The dataset's contents and structure should be clearly described. Relevant information includes:

- What constitutes an individual instance or record?
- How many observations are included?
- Which populations, environments, or time periods are represented?
- Are labels or annotations provided?
- Does the dataset contain sensitive or protected information?

### Collection process

The process used to create the dataset should be documented. This includes:

- who collected the data;
- what instruments, systems, or procedures were used;
- the sampling strategy;
- whether informed consent was obtained;
- whether participants received compensation;
- and any known limitations of the collection process.

### Preprocessing and cleaning

All modifications applied to the original data should be recorded. Documentation should describe:

- which records were removed;
- whether missing values were imputed;
- whether variables were normalized or transformed;
- whether labels were reviewed or adjudicated;
- whether identifiers were removed or anonymized;
- and any other cleaning procedures.

### Intended and discouraged uses

Dataset creators should clearly state appropriate and inappropriate applications. Documentation should identify:

- intended tasks;
- recommended users;
- known limitations;
- prohibited or discouraged uses;
- contexts where performance may be unreliable.

### Distribution and licensing

Users should understand how the dataset can be accessed and shared. Relevant information includes:

- access procedures;
- applicable licenses;
- redistribution permissions;
- commercial-use restrictions;
- requirements for attribution;
- and restrictions related to privacy or security.

*Example # 56: Medical imaging dataset*

A research group downloads a publicly available medical imaging dataset to train a disease-detection AI model. The dataset documentation states:

- **Access procedure:** Requires approval from the data custodian and completion of a data-use agreement.
- **License:** Allows academic research but prohibits commercial use.
- **Attribution:** Users must cite the original dataset publication.
- **Privacy restriction:** Images cannot be redistributed because they contain sensitive patient information.

Without this information, another organization might **download and redistribute the images publicly**, use them in a **commercial product**, or fail to provide proper attribution, potentially **violating privacy** regulations, **intellectual property rights**, or the dataset owner's terms.

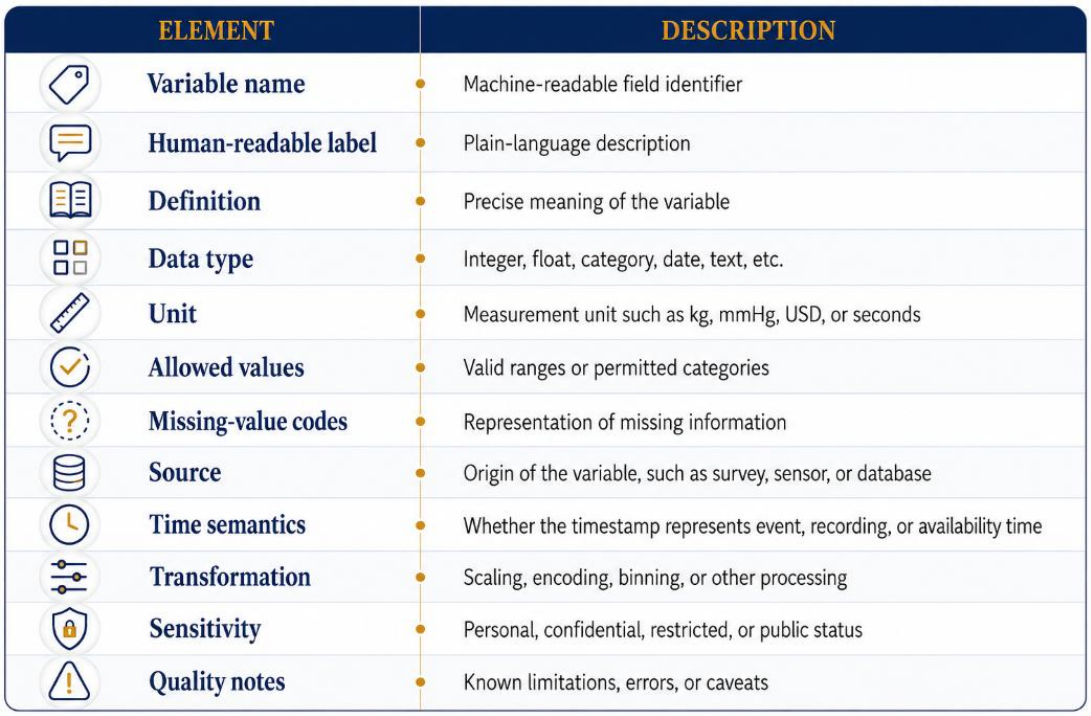
### Maintenance

Long-term responsibility for the dataset should be specified. Documentation should identify:

- dataset owners or maintainers;
- procedures for reporting errors;
- update schedules;
- withdrawal procedures;
- versioning practices;
- and methods for tracking revisions.

## 2.7.2 Data Dictionaries

A data dictionary provides **field-level documentation** describing the **meaning**, **structure**, and **interpretation** of individual variables. A useful data dictionary should include the items listed in Figure 18.



| ELEMENT | DESCRIPTION |
|---|---|
| Variable name | Machine-readable field identifier |
| Human-readable label | Plain-language description |
| Definition | Precise meaning of the variable |
| Data type | Integer, float, category, date, text, etc. |
| Unit | Measurement unit such as kg, mmHg, USD, or seconds |
| Allowed values | Valid ranges or permitted categories |
| Missing-value codes | Representation of missing information |
| Source | Origin of the variable, such as survey, sensor, or database |
| Time semantics | Whether the timestamp represents event, recording, or availability time |
| Transformation | Scaling, encoding, binning, or other processing |
| Sensitivity | Personal, confidential, restricted, or public status |
| Quality notes | Known limitations, errors, or caveats |

**Figure 18: Core elements of a data dictionary.** The figure summarizes the field-level metadata required to document the meaning, structure, provenance, processing, and appropriate use of dataset variables. For each variable, a data dictionary should record its machine-readable name, human-readable label, formal definition, data type, unit, allowed values, missing-value codes, source, temporal semantics, transformations, sensitivity classification, and known quality limitations.

### 2.7.3 Provenance and data lineage

**Data provenance** describes the **origin of data** and the **processes** used to generate them. **Data lineage** records **how data evolve** as they move through processing pipelines. A lineage chain is represented in **Figure 19**.

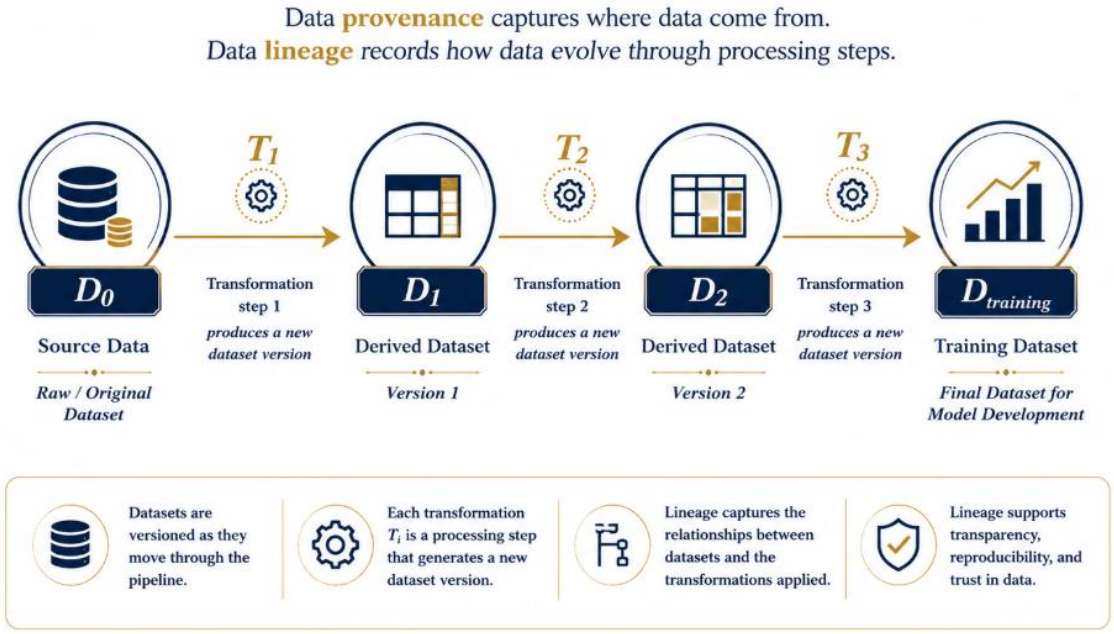



**Figure 19: Data lineage chain from source data to model-ready dataset.** The figure illustrates how data provenance and lineage can be represented as a sequential chain of dataset versions connected by explicit transformations. Starting from the original source dataset, each processing step produces a new derived dataset version, culminating in the final training dataset used for model development. This representation emphasizes traceability, reproducibility, and transparency by making the relationships between data sources, transformation steps, and downstream analytical artifacts explicit.

For every transformation, organizations should record:

- input dataset version;
- code or software version;
- processing parameters;
- output dataset version;
- execution timestamp;
- responsible person or system;
- and validation results.

#### Why provenance matters

Provenance supports:

- reproducibility of analyses;
- debugging of errors;
- auditing and accountability;
- correction of faulty processes;
- rollback to earlier versions;
- regulatory compliance;
- and evaluation of dataset suitability.

A trained model should be **traceable back** to the **exact dataset version**, **preprocessing steps**, and **code** used during development.

### 2.7.4 Consent documentation

Consent should not be represented merely as a binary indicator such as **consent obtained: Yes/No**. **Meaningful consent documentation** requires **understanding** what **participants agreed to** and under what conditions. Important questions include:

- What information were participants provided?
- Was participation voluntary?
- What purposes were explained?
- Did consent include data linkage?
- Did consent permit commercial use?
- Did consent cover international data transfer?
- Could participants withdraw consent?
- What happens to derived models or analyses after withdrawal?
- Were vulnerable populations involved? Was consent broad or specific?

In health-related applications, **guidance from organizations** such as the WHO **emphasizes** the **importance of autonomy**, **privacy**, **confidentiality**, and **informed consent** as essential components of **responsible AI governance**.

### 2.7.5 Licensing

A data license **specifies the conditions** under which a dataset **may be accessed**, **used**, **modified**, or **redistributed**. Possible restrictions may address:

- commercial use;
- modification;
- redistribution;
- attribution requirements;
- creation of derivative works;
- geographic restrictions;
- research-only access;
- attempts at re-identification;
- and uses involving surveillance or discrimination.

### 2.7.6 Copyright and privacy are different

Legal permission to access a dataset **does not automatically mean** that **every possible use is acceptable**.

A dataset **may have an open license** while **still containing personal information** that **restricts certain applications**. Conversely, a dataset **may contain de-identified information** while **remaining subject** to **copyright**, **contractual**, or **institutional restrictions**.

Privacy obligations and intellectual property rights must therefore be considered separately.

### 2.7.7 License compatibility

When **combining multiple datasets**, **license compatibility** must be evaluated carefully.

Datasets with **incompatible licenses** may **prevent redistribution**, **publication**, or **deployment of systems** built from the combined data.

The dataset license, applicable version, and any usage restrictions **should always be recorded** alongside the dataset artifact to **ensure future users** understand the permitted conditions of use.

*Example # 57: AI model to predict Alzheimer's Disease*

A university creates an AI model to predict **early signs of Alzheimer's disease** using MRI scans collected from 5,000 participants.

*Consent issue*

The **consent documentation only** includes the following: Consent obtained: Yes. However, the original consent form specified that participants agreed only to:

- academic research use;
- analysis of brain imaging data;
- no commercial use;
- no transfer of data outside the country.

Years later, a technology company wants to use the dataset to develop a commercial diagnostic product and share the trained model internationally.

Although consent was recorded as Yes, the **proposed use** may **violate the participants' actual agreement** because:

- commercial use was not permitted;
- international data transfer was not approved;
- participants were not informed about future AI model development.

A detailed consent record allows researchers to determine whether the intended use is ethically and legally acceptable.

*License issue*

The MRI dataset is released under a **research-only license** that allows access and analysis but prohibits:

- commercial deployment;
- redistribution of raw images;
- attempts to identify participants.

A researcher combines this dataset with **another openly licensed dataset** and publishes the combined dataset online. This **may violate** the original license because:

- the first dataset does not permit redistribution;
- the combined dataset inherits restrictions;

- downstream users cannot assume that all open datasets have compatible permissions.

*Key lessons*

Consent explains **what participants agreed to**, while licensing defines **what users are legally permitted to do with the data**. Both must be documented because access to data does not automatically mean unrestricted ethical or legal use.

# Governance and Ethics

**Technical excellence** alone is **insufficient for responsible** data science and artificial intelligence. The **collection**, **management**, and **use of data** must also be guided by **sound governance** and **ethical principles** that **protect individual rights**, **promote accountability**, and ensure that data-driven systems **serve legitimate societal objectives**. Effective governance establishes clear responsibilities, policies, and controls for managing data throughout its lifecycle, while ethical practice requires careful consideration of privacy, consent, fairness, transparency, and the potential impacts of automated decision-making. We discuss the **fundamental principles of data governance and ethics** and examine practical approaches for ensuring that **data are managed responsibly**, **lawfully**, and in ways that **inspire public trust**.

## 2.8.1 Data governance

Data governance is the **framework of policies**, **roles**, **processes**, and **technical controls** through which **data are managed**, **protected**, and **used throughout their lifecycle**. An effective governance framework addresses fundamental questions such as:

- Who owns or serves as steward of the data?
- Who is authorized to access the data?
- For what purposes may the data be used?
- Who approves new or secondary uses?
- How long should the data be retained?
- How are data quality issues corrected?
- How are security or privacy incidents managed?
- How are model outputs monitored and audited?
- Who is ultimately accountable for decisions involving the data?

Governance extends **beyond written policies**. It **must be implemented** through **operational procedures**, **technical safeguards**, **monitoring mechanisms**, and clearly **assigned responsibilities**.

## 2.8.2 Privacy

Privacy concerns an **individual's ability to control**, **understand**, and be **protected** from **inappropriate collection**, **processing**, **inference**, **sharing**, or **disclosure of information** about them. Privacy risks extend beyond direct identification and may **arise even** when **explicit identifiers have been removed**. Common privacy risks include:

- direct identification;
- re-identification of supposedly anonymous individuals;
- linkage attacks using multiple datasets;
- membership inference attacks;
- reconstruction of sensitive information;
- inference of confidential attributes;
- unauthorized secondary use of data;
- surveillance;
- and indefinite or unnecessary data retention.

Protecting privacy requires considering the **entire data lifecycle** rather than focusing solely on data collection.

### 2.8.3 De-identification is not absolute

Removing names, identification numbers, or other direct identifiers **does not necessarily** make a dataset **anonymous**.

Combinations of **seemingly innocuous variables** – such as **age**, **geographic location**, **dates**, or **rare characteristics** – may **uniquely identify** individuals when linked with external information.

Consequently, de-identification should be viewed as a **risk-reduction strategy** rather than a **guarantee of anonymity**.

### 2.8.4 Privacy throughout the data lifecycle

Privacy protection should be **incorporated into every stage** of data management. Common safeguards include:

- collecting only necessary information;
- limiting data use to specified purposes;
- restricting access according to user roles;

- encrypting stored and transmitted data;
- using secure computing environments;
- publishing aggregated rather than individual-level data;
- applying differential privacy where appropriate;
- adopting federated or distributed learning approaches;
- establishing retention limits;
- maintaining audit logs; and
- implementing incident response procedures.

International guidance, including **principles developed** by the **OECD**[253,254], emphasizes **lawful processing**, **purpose limitation**, **data minimization**, **transparency**, **security**, **respect** for **individual rights**, and **accountability**.

### 2.8.5 Data minimization

Data minimization is the **principle** that **organizations should collect**, **retain**, and **process only** the information **reasonably necessary** to achieve a **clearly defined** and **legitimate purpose**. A useful decision criterion is:

$$\text{Collect } X_j \quad \text{only if} \quad \text{Benefit}(X_j) > \text{Risk}(X_j)$$

taking into account the **intended use** and **available safeguards**. Questions supporting data minimization include:

- Is this variable genuinely necessary?
- Could a less sensitive alternative provide the required information?
- Is long-term retention required?
- Can access be restricted to authorized users?
- Must raw values be preserved?
- Is the variable needed for fairness auditing even if it is excluded from prediction?

[253] **OECD (Organization for Economic Co-operation and Development):** An international organization that develops evidence-based policies and guidelines for economic, social, and technological governance, including widely adopted principles for trustworthy AI and data governance. Example: The **OECD AI Principles** recommend that healthcare AI systems process patient data lawfully, minimize data collection, ensure transparency, protect privacy, and maintain accountability throughout the AI lifecycle.

[254] Organization for Economic Co-operation and Development (OECD), *OECD Principles on Artificial Intelligence*, Paris: OECD Publishing, 2019.

Data minimization **reduces privacy risks** while **encouraging efficient** and **responsible data management**.

*Example # 58: Hospital readmission study*

A hospital develops an AI model to predict whether a patient will be **readmitted within 30 days**. The research team initially plans to collect the variables shown in **Table 25**.

**Table 25: Variable-necessity assessment for clinical readmission modeling.**

| Variable | Needed | Reason |
|---|---|---|
| Age | ✓ | Strong predictor of readmission |
| Previous hospital admissions | ✓ | Clinically relevant |
| Current diagnoses | ✓ | Required for prediction |
| Home address | ✗ | Postal code provides sufficient geographic information |
| National ID number | ✗ | Not needed for prediction |
| Full medical notes | ✗ | Key diagnoses are already available as structured variables |
| Ethnicity | ✓ *(restricted access)* | Used only to assess model fairness, not for prediction |

✓ Included for prediction ✗ Excluded from prediction ✓ Included with restricted access (non-predictive use only)

**Instead** of collecting **full addresses**, **national ID** numbers, and complete **clinical notes**, the team **retains only age**, previous **hospital admissions**, **current diagnoses**, **postal code**, and **ethnicity** (with restricted access solely for fairness auditing), because **these variables are sufficient** for the study while reducing privacy risks.

### 2.8.6 Sensitive attributes

Sensitive attributes are **variables** whose **misuse may expose individuals** to **discrimination**, **privacy violations**, or other **forms of harm**. Examples include:

- race or ethnicity;
- religion;

- health information;
- disability;
- gender;
- orientation;
- political beliefs;
- precise geographic location;
- biometric information;
- financial information;
- and immigration status.

The handling of sensitive attributes requires **enhanced safeguards** and **careful governance**.

### Removing sensitive attributes does not eliminate bias

**Excluding a sensitive variable** from a predictive model **does not necessarily prevent discriminatory outcomes**. Other variables **may function as proxies** for the omitted attribute: $X_{\text{proxy}} \leftrightarrow A_{\text{sensitive}}$. Examples of proxy variables include:

- postal codes;
- schools attended;
- preferred language;
- purchasing behavior;
- occupation;
- and healthcare utilization.

Consequently, **bias may persist** even when **sensitive variables are excluded** from model inputs.

*Example # 59: Bank approving loans*

A bank develops an AI model to predict whether applicants should be **approved for a home loan**. To avoid discrimination, the bank **removes race** from the training data. However, the model still uses:

- Postal code
- Household income
- Occupation

Many neighborhoods are **strongly associated** with particular **racial** or **ethnic groups** due to historical residential patterns. As a result, **postal code acts as a proxy for race**.

The model **consistently approves** applicants from **affluent postal codes** while **rejecting applicants** from **disadvantaged neighborhoods**, even when applicants have similar financial profiles.

Although **race is not an input variable**, the model still produces different outcomes because it indirectly learns information about race from correlated variables.

### Sensitive attributes for fairness auditing

Although **sensitive attributes may be inappropriate** as predictive features in some applications, they are **often essential** for evaluating fairness. Without **subgroup information**, it may be impossible to **identify systematic disparities** in model performance. Governance frameworks should therefore distinguish among:

- use for prediction;
- use for fairness evaluation;
- use for ongoing monitoring; and
- access, retention, and security controls.

*Example # 60: Hospital readmission study*

A hospital develops an AI model to predict which patients are at **high risk of hospital readmission**. To reduce the possibility of discrimination, the model **does not use ethnicity as a predictive feature**. After deployment, the hospital evaluates the model separately for different ethnic groups (**Table 26**).

**Table 26: Prediction accuracy across ethnic groups.**

| Ethnic Group | Prediction Accuracy |
|---|---|
| Group A | 91% |
| Group B | 90% |
| Group C | 74% |

The results show that the model performs substantially worse for **Group C**.

Without recording **ethnicity** for fairness evaluation, this **disparity would remain undetected**, and the hospital could unknowingly provide less reliable predictions for one group of patients.

To address this, the hospital:

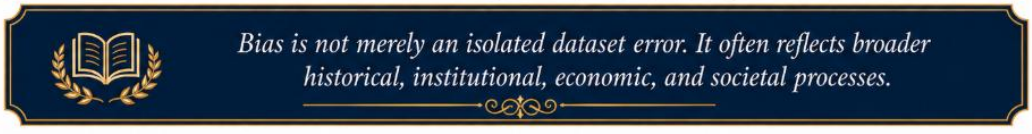
- **Excludes ethnicity from prediction** to avoid direct use in decision-making.
- **Retains ethnicity with restricted access** for fairness evaluation and ongoing monitoring.
- **Applies strict access and security controls** so that only authorized personnel can use the information for auditing.

### 2.8.7 Bias

Bias may **enter a data-driven system** at **multiple stages** of its **lifecycle**, from **data collection** to **deployment**. Because these sources often interact, addressing bias requires a **comprehensive understanding** of the entire system rather than focusing on individual variables (Table 27).

**Table 27: Types of bias in AI systems.** Seven forms of bias may arise across data collection, modeling, evaluation, deployment, and feedback processes, reflecting both technical and broader societal influences.

| Bias Type | Description |
|---|---|
| Historical Bias | Existing social, economic, or institutional inequalities are reflected in the collected data. |
| Representation Bias | Important populations, environments, or scenarios are absent or substantially underrepresented. |
| Measurement Bias | Variables, instruments, or labels measure different groups with unequal accuracy or reliability. |
| Aggregation Bias | A single predictive model is applied to populations with different underlying relationships between predictors and outcomes. |
| Evaluation Bias | Benchmarks or evaluation procedures fail to reflect the intended deployment environment or the harms associated with prediction errors. |
| Deployment Bias | A system is used in ways that differ from its intended purpose or from the conditions under which it was validated. |
| Feedback Bias | Model predictions influence future data collection or decision-making, thereby reinforcing existing patterns and potentially amplifying inequities over time. |



*Example # 61: Real-world examples of bias*

**Historical bias:** A hospital trains an AI model using historical treatment data. In the past, patients from low-income communities were referred to specialists less often because of limited access to care. The model learns these historical patterns and predicts fewer specialist referrals for similar patients, perpetuating existing inequalities.

**Representation bias:** An AI system for detecting skin cancer is trained primarily on images of light skin. When deployed, it performs well for light-skinned patients but poorly for dark-skinned patients because those cases were underrepresented in the training data.

**Measurement bias:** A wearable device estimates heart rate accurately for younger adults but is less accurate for older adults due to age-related physiological differences. As a result, the recorded measurements are systematically less reliable for one group.

**Aggregation bias:** A single model predicts kidney disease risk using data from adults aged $18 - 90$. Because risk factors differ substantially between young adults and older adults, one model performs poorly for both groups compared with age-specific models.

**Evaluation bias:** An AI model for pneumonia detection is evaluated only on high-quality chest X-rays from a large urban hospital. When deployed in rural clinics with older imaging equipment, its accuracy drops substantially because the evaluation data did not represent the intended deployment environment.

**Deployment bias:** An AI model validated to assist radiologists in identifying lung nodules is later used as the sole basis for clinical decisions without expert review. The system is being used outside the conditions for which it was validated, increasing the risk of diagnostic errors.

**Feedback bias:** A predictive policing system identifies certain neighborhoods as high risk, leading to increased police patrols in those areas. The additional patrols detect more incidents, creating more data that reinforces the model's

belief that these neighborhoods are high risk, even if the underlying crime rate has not changed.

### 2.8.8 Fairness

Fairness is a **context-dependent concept** that **cannot be** adequately **characterized** by a **single universal metric**. Different applications involve different **stakeholders**, **risks**, and **ethical priorities**, making the choice of fairness criterion dependent on the decision being supported. For **binary classification**, commonly used group fairness metrics include:

- **Demographic parity:** Predicted positive rates are equal across groups.

$$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = b)$$

- **Equal opportunity:** True positive rates are equal across groups.

$$P(\hat{Y} = 1|Y = 1, A = a) = P(\hat{Y} = 1|Y = 1, A = b)$$

- **Equalized odds:** Equalized odds require equality of both true positive rates and false positive rates across demographic groups.
- **Predictive parity:** The probability that a positive prediction is correct is equal across groups.

$$P(Y = 1|\hat{Y} = 1, A = a) = P(Y = 1|\hat{Y} = 1, A = b)$$

These fairness criteria **may be mutually incompatible**, particularly when **outcome prevalence differs between populations**. Consequently, selecting an appropriate fairness metric requires consideration of:

- the decision being made;
- the potential harms of prediction errors;
- applicable legal and ethical requirements;
- stakeholder values; and
- available opportunities for intervention.

Fairness evaluations **should also consider intersectional groups**, as **aggregate demographic categories** may conceal **substantial disparities**.

The Gender Shades[255] study provides a well-known example of this phenomenon.

*Example # 62: Choosing an appropriate fairness metric*

A hospital develops an AI model to identify patients at **high risk of sepsis**. The model is used for two demographic groups:

- **Group A:** Sepsis prevalence $= 20\%$
- **Group B:** Sepsis prevalence $= 5\%$

The hospital considers different fairness criteria:

- **Demographic parity** would require the model to flag **the same proportion of patients** in both groups. Because Group A has a much higher true prevalence of sepsis, this could lead to *over-testing low-risk patients in Group B* or *under-detecting high-risk patients in Group A*.
- **Equal opportunity** would require the model to identify **the same proportion of patients who truly develop sepsis** in both groups (equal true positive rates). This is desirable because missing a patient with sepsis can have serious clinical consequences.
- **Predictive parity** would require that, among patients predicted to have sepsis, **the probability of actually having sepsis** is the same in both groups. Achieving this may require using different decision thresholds, which can make equal opportunity impossible when disease prevalence differs.

Because **Group A has a higher prevalence of sepsis than Group B**, it is generally **impossible to satisfy all three fairness criteria simultaneously**. The hospital therefore chooses **equal opportunity**, prioritizing equal detection of true sepsis cases because **false negatives** (missed cases) **pose greater harm** than **false positives** (additional diagnostic tests)**.**

### 2.8.9 Consent and secondary use

[255] Joy Buolamwini and Timnit Gebru, *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*, *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, PMLR 81:77-91, 2018.

Data collected **for one purpose** are **often valuable for other applications** that were **not anticipated** during the **original collection process**. Examples include:

- educational records used for disciplinary prediction;
- health records used for commercial marketing;
- social media content used for psychological profiling;
- and location histories used for surveillance.

The technical ability to access data **does not by itself justify new uses**. Before approving secondary use, **organizations should evaluate**:

- participants' original expectations;
- compatibility with the original purpose;
- potential risks of harm;
- applicable legal authority;
- the scope of informed consent;
- community expectations;
- oversight requirements;
- and whether participants can withdraw consent.

### 2.8.10 Institutional approval

Projects involving **sensitive or high-risk data** frequently **require review** by appropriate **oversight bodies**. Depending on the context, these may include:

- institutional review boards or research ethics committees;
- data protection offices;
- information security teams;
- legal counsel;
- clinical governance committees;
- community advisory groups;
- and regulatory authorities.

Review should consider:

- scientific necessity;
- expected benefits and risks;
- informed consent;

- privacy protections;
- vulnerable populations;
- access controls;
- data retention policies;
- data sharing practices;
- incident response procedures;
- and planned model deployment.

**Ethical approval** is an **important safeguard** but **does not replace** continuous **technical monitoring** and **governance** throughout a **project's lifecycle**.

### 2.8.11 Governance roles

Effective governance depends on clearly defined responsibilities:

- **Data owner:** Holds formal authority and ultimate responsibility for the data resource.
- **Data steward:** Maintains data definitions, quality standards, documentation, metadata, and access policies.
- **Data custodian:** Manages the technical infrastructure supporting data storage, security, backup, and access.
- **Data producer:** Creates, measures, or collects the original observations.
- **Data consumer:** Uses the data for research, analysis, reporting, or predictive modeling.
- **Model owner:** Assumes responsibility for the development, deployment, monitoring, and operational performance of predictive models.
- **Ethics or Oversight body:** Reviews projects involving significant ethical, legal, or societal considerations, including privacy, consent, fairness, public interest, and protection of individual rights.

Clearly assigning these responsibilities **promotes accountability** and helps **prevent a common governance failure** in which many individuals use the data, but **no one is explicitly responsible** for its quality, security, or appropriate use.

# How a “Good” Dataset Produces a Misleading AI System

The **quality** of an artificial intelligence system **depends not only** on the **quality** of its **data** but also on whether the **data accurately represent the problem** the system is **intended to solve**. A dataset may satisfy conventional quality criteria (such as completeness, consistency, accuracy, and balance**) yet still produce misleading or unfair predictions** if it measures the **wrong outcome**, **omits important populations**, or **encodes historical biases**. In such cases, the model faithfully learns patterns present in the data while **failing to capture the underlying phenomenon** of interest. We explore how **seemingly high-quality** datasets can lead to **unreliable AI systems** and present a systematic framework for evaluating whether a **dataset is genuinely suitable** for its intended application.

## 2.9.1 The misleading dataset pattern

A dataset may appear to be of exceptionally high quality because it is:

- large;
- nearly complete;
- well structured and consistently formatted;
- balanced across classes;
- consistently labeled;
- easily accessible;
- and highly predictive.

Despite these **desirable characteristics**, the dataset **may still be fundamentally unsuitable** if it measures the **wrong phenomenon**.

A useful way to understand this problem is through the following sequence:

True Construct → Institutional Process → Recorded Proxy → Training Target

The **true construct** is the concept that the **project seeks to predict**, but it is often **difficult or impossible to observe** directly. Consequently, organizations record a measurable proxy instead.

Machine learning models **learn relationships** involving the **recorded proxy**, **not necessarily** the **underlying construct**. If the proxy **systematically differs** from the true concept, the resulting model may achieve excellent predictive performance while **making decisions** that are **misleading or unfair**.

*Example # 63: Hospital admission risk*

Consider the setup depicted in **Table 28**.

**Table 28: From true healthcare need to the model training target.**

| Stage | Example |
|---|---|
| True construct | A patient's **true need for healthcare** after discharge. |
| Institutional process | Hospitals decide whether to **readmit** patients based on clinical judgment, bed availability, insurance policies, and local practices. |
| Recorded proxy | Whether the patient was **readmitted within 30 days** (Yes/No). |
| Training target | The AI model is trained to predict **30-day readmission**, treating it as the outcome of interest. |

However, **30-day readmission is only a proxy for healthcare need**. A patient who needs hospital care but cannot return because of financial barriers, lack of transportation, or limited hospital capacity will be recorded as **not readmitted**, even though the true construct (need for care) is high.

The AI model learns to predict the **recorded proxy (readmission)** rather than the **true construct (healthcare need)**. If the proxy is influenced by institutional or societal factors, the model may inherit those biases.

*Example # 64: Predicting criminal recidivism[256] risk*

Consider the setup depicted in **Table 29**.

[256] **Recidivism:** The occurrence of a person **committing a new offense** or **returning** to the **criminal justice system** after a **previous conviction, sanction, or release**. Example: A person released after serving a sentence for burglary is arrested again for a new burglary offense within two years; this new offense is considered recidivism.

**Table 29: From true offending risk to the model training target.**

| Stage | Example |
|---|---|
| True construct | An individual's **actual likelihood** of committing a future harmful offense after release. |
| Institutional process | The criminal justice system determines whether an individual is **detected, investigated, arrested, charged, and convicted** based on policing practices, reporting behavior, resource allocation, judicial decisions, and access to legal support. |
| Recorded proxy | Whether the individual has a **new arrest or conviction** within a **defined period** (e.g., two years after release). |
| Training target | An AI model is trained to predict **future arrest or reconviction risk** and uses this as a measure of future offending risk. |

However, **arrest or conviction is only a proxy for actual offending behavior**. A person may commit an offense but not be detected or arrested, while another person may have a higher probability of arrest because of differences in policing intensity, neighborhood surveillance, reporting patterns, or institutional practices.

For example:

- Person A and Person B may have similar underlying likelihood of reoffending.
- Person A lives in an area with frequent police patrols and is more likely to be detected and arrested.
- Person B lives in an area with less police presence and is less likely to appear in official records.

The AI system may learn that Person A's characteristics are associated with **higher risk**, not because Person A has a higher true likelihood of offending, but because the **recorded outcome reflects institutional processes**.

The model predicts the **recorded proxy (arrest/conviction)** rather than the **true construct (actual future harmful behavior)**. When the proxy is shaped

by social, institutional, or procedural factors, the model may **reproduce those patterns** unless the **target definition** and **evaluation** process are **carefully examined**.

### 2.9.2 Healthcare need versus healthcare cost

Consider a healthcare system seeking to identify patients who require additional medical care. The desired prediction target is $Y^* = \text{Health Need}$.

Because health need **cannot be measured directly**, developers instead choose future healthcare expenditure as training target:

$$Y = \text{Future Healthcare Cost}$$

However, healthcare expenditure depends on many factors beyond medical need:

$$Y = f(Y^*, \text{Access}, \text{Insurance}, \text{Provider}, \text{Behavior}, \text{Historical Inequality})$$

Consequently, two individuals with identical health needs may incur very different healthcare costs because of differences in insurance coverage, access to healthcare services, provider practices, or longstanding structural inequities.

A model trained to predict healthcare expenditure may therefore **learn historical patterns** of **healthcare spending** rather than actual healthcare need.

As demonstrated by Obermeyer and colleagues[257], a widely used population-health management algorithm **systematically underestimated** the **needs of Black patients**. At the same predicted risk score, **Black patients were generally sicker** than White patients because **healthcare cost had been used as a proxy** for healthcare need. The model accurately predicted spending while reproducing historical disparities in access to care.

### 2.9.3 A dataset suitability audit

[257] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan, *Dissecting racial bias in an algorithm used to manage the health of populations*, Science, 366(6464), pp 447-453, 2019.

Before adopting a dataset for analysis or model development, its **suitability should be evaluated** systematically. The following questions provide a **practical framework** for assessing whether the dataset appropriately represents the intended problem.

### Construct validity

Evaluate the **relationship between** the **intended concept** and the **recorded target**. Key questions include:

- What real-world construct is the project intended to predict?
- What does the available label actually measure?
- Is the label a direct measurement or a proxy?
- If it is a proxy, how might it differ from the intended construct?
- Could the proxy behave differently across demographic or operational groups?

### Sampling

Assess whether the dataset **adequately represents the target population**. Questions include:

- Who is included in the dataset?
- Who is absent or underrepresented?
- What sampling process generated the observations?
- Does inclusion depend on access, participation, or the outcome itself?
- Does the dataset reflect the population in which the model will ultimately be deployed?

### Time

Verify that the **temporal characteristics** of the data are **appropriate**. Important questions include:

- Will every predictor be available at the intended prediction time?
- Have variable definitions or coding practices changed over time?
- Does the dataset represent the future operating environment?
- Could temporal shifts affect model performance?

### Measurement

Examine **how variables and labels were obtained**. Questions include:

- Which instruments or measurement procedures were used?
- Who performed labeling or annotation?
- Are measurement errors documented?
- Do error rates differ across population subgroups?
- Are important variables measured consistently over time?

### Governance

Evaluate whether the **dataset satisfies legal**, **ethical**, and **organizational requirements**. Consider:

- Was the dataset collected lawfully?
- Was data collection ethically justified?
- Does informed consent cover the proposed application?
- Are licensing conditions clearly documented?
- Are access controls and responsibilities appropriately defined?

### Evaluation

Finally, evaluate whether the **assessment** methodology **accurately reflects** the **intended deployment setting**. Questions include:

- Is the test dataset independent of the training data?
- Does it represent the target deployment population?
- Are performance metrics reported separately for relevant subgroups?
- Has model calibration been evaluated?
- Is performance compared with meaningful baseline methods?

A dataset should **not be judged** solely by its **size**, **completeness**, or **predictive performance**. Rather, its suitability depends on whether it **faithfully represents** the underlying **construct**, **population**, **measurement process**, and **deployment environment** associated with the problem the model is intended to solve.

# Recommended Operational Framework

A **disciplined operational** framework is needed to determine whether data are **appropriate for use before they influence** research, development, or deployment decisions. Dataset acceptance should therefore be treated as a **structured evaluation process** that examines **purpose**, **provenance**, **quality**, **leakage risks**, **governance requirements**, and **suitability for the intended application**. A practical framework is presented for assessing datasets helping organizations **identify limitations early**, **prevent avoidable failures**, and ensure that data-driven systems are **built on a responsible foundation**.

## 2.10.1 Data acceptance gate

Before a dataset is incorporated into model development, it should undergo a **structured acceptance review** to determine whether it is **appropriate**, **reliable**, and **responsibly usable**. A dataset should pass **six essential evaluation gates** before entering the modeling pipeline (**Figure 20**).

## 2.10.2 Minimum dataset artifacts

Every serious AI project should maintain:

- raw-data inventory;
- source and provenance register;
- data dictionary;
- schema;
- quality report;
- missingness report;
- duplicate and entity-resolution report;
- label-definition document;
- split specification;
- leakage audit;
- datasheet or data card;
- consent and licensing record;
- access-control record;
- version history;

- maintenance and incident plan.




**Figure 20: Dataset acceptance gate for model development.** The figure presents a six-gate operational framework for deciding whether a dataset is suitable to enter model development.

# Three Illustrative Case Studies

We now discuss three case studies utilizing the concepts, principles, and practices studied in this module.

## 2.11.1 Gender Shades and Intersectional Representation Failure

### Background

Commercial facial-analysis systems had achieved impressive aggregate performance and were increasingly being used in consumer, security, and organizational applications. Aggregate accuracy, however, could conceal whether systems worked equally well for different demographic groups.

Buolamwini and Gebru[258] constructed the Pilot Parliaments Benchmark, designed to be more balanced across gender presentation and skin type, and evaluated three commercial gender-classification systems. They found substantial intersectional disparities. Error rates reached 34.7% for darker-skinned women, whereas the maximum error rate for lighter-skinned men was 0.8%. They also reported that prominent existing face datasets were heavily skewed toward lighter-skinned subjects.

### Why the original datasets could appear good

The source datasets had several characteristics commonly associated with quality:

- thousands of images;
- labeled demographic categories;
- recognizable faces;
- public research use;
- established benchmark status.

Yet **aggregate size** did not **ensure balanced coverage**. Suppose a test set contains four intersectional groups:

$$G = \{\text{lighter men}, \text{lighter women}, \text{darker men}, \text{darker women}\}$$

Overall error is:

[258] Joy Buolamwini and Timnit Gebru, *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*, *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, PMLR 81:77-91, 2018.

$$E_{\text{overall}} = \sum_{g \in G} P(G = g) E_g$$

If the **easiest group dominates** the dataset, **overall error can be low** even when **one minority group** has an **unacceptable error rate**.

### Data-quality diagnosis

- **Representativeness failure:** The benchmark composition did not adequately reflect demographic diversity.
- **Aggregation failure:** Reporting performance by gender alone or skin type alone could obscure the worst intersection.
- **Measurement and labeling:** Gender presentation and skin-type categorization involve operational and social assumptions that require careful documentation.
- **Evaluation bias:** The benchmark was not constructed to reveal performance disparities relevant to broad deployment.

### Why cleaning alone would not solve the problem

There **may be no** corrupted image or malformed label **to repair**. The problem lies in:

- who was sampled;
- the proportions represented;
- which subgroup metrics were reported;
- the social meaning of categories;
- the intended deployment claim.

**Traditional cleaning** can produce a **perfectly formatted** but **still inequitable dataset**.

### Redesigned data strategy

- **Define the deployment population**
  Document expected variation in
  - skin tone;
  - age;
  - gender presentation;
  - lighting;

- camera type;
- pose;
- occlusion;
- geography.

- **Establish subgroup sampling targets**
  - For each relevant group $g$, define a minimum evaluation size: $n_g \geq n_{\min}$.
  - The test set should support precise subgroup estimates, even when natural prevalence is unequal.

- **Document annotation**

  Record
  - category definitions;
  - annotator instructions;
  - disagreement;
  - self-identification versus external labeling;
  - uncertainty.

- **Report disaggregated metrics**

  For every group
  - error rate;
  - sensitivity;
  - specificity;
  - confidence interval;
  - calibration;
  - failure examples.

- **Monitor post-deployment performance**
  - New devices, lighting environments, and populations may introduce further shifts.

### Ethical significance

Performance disparities in face analysis are consequential because such systems can influence:

- identity verification;
- access control;
- employment workflows;

- surveillance;
- law enforcement;
- consumer services.

An error may not be a mere technical inconvenience. It **may create exclusion**, **suspicion**, **humiliation**, or **denial of service**.

#### Lessons

- Large sample size does not guarantee representation.
- Aggregate accuracy can conceal severe subgroup failure.
- Intersectional evaluation is often necessary.
- Dataset composition is an ethical and technical design decision.
- Fairness cannot be repaired only after training.
- Documentation should explain how demographic categories were constructed.
- Data collection must be aligned with deployment claims.

This case establishes the **inevitability of dataset governance**: the **model's disparity** could not be understood without **investigating who was represented**, how **groups were defined**, and how **evaluation was aggregated**.

### 2.11.2 Racial Bias from a Healthcare-Cost Proxy

#### Background

Population-health programs use risk scores to identify patients who may benefit from additional care-management resources. Because direct health need is difficult to measure, developers may choose future healthcare expenditure as a target.

Obermeyer and colleagues examined a widely used commercial algorithm and found that, **at the same risk score**, **Black patients** were **considerably sicker** than **White patients**. The underlying mechanism was the use of **healthcare cost** as a **proxy for health need**. Because less money was historically spent on Black patients with comparable illness, the model **systematically underestimated their need**.

### The mathematical structure of the error

Let $H$ = true health need, $C$ = future healthcare cost, $A$ = group membership. The desired model is $f(X) \approx H$. The implemented model is $f(X) \approx C$. But

$$C = g(H, A, \text{access}, \text{insurance}, \text{provider behavior})$$

where $A$ could be **race**, access could be the **ability to obtain healthcare services**, insurance be the **coverage** and **reimbursement factors**, and provider behavior may represent **clinical decisions**, **referral patterns**, **treatment practices**.

Even if the model predicts $C$ accurately, $\hat{C} \approx C$, it does not fellow that $\hat{C} \approx H$. The target is statistically learnable but conceptually misaligned.

### Why the dataset could appear high quality

The administrative dataset may have been:

- large;
- longitudinal;
- consistently coded;
- complete for billing;
- directly connected to healthcare operations;
- predictive of future cost.

From an engineering perspective, it may have been easier to obtain than a carefully constructed measure of illness burden. The failure therefore was not conventional data corruption. It was a **construct and proxy failure**.

### Bias pathway

A simplified pathway is depicted in **Figure 21**.

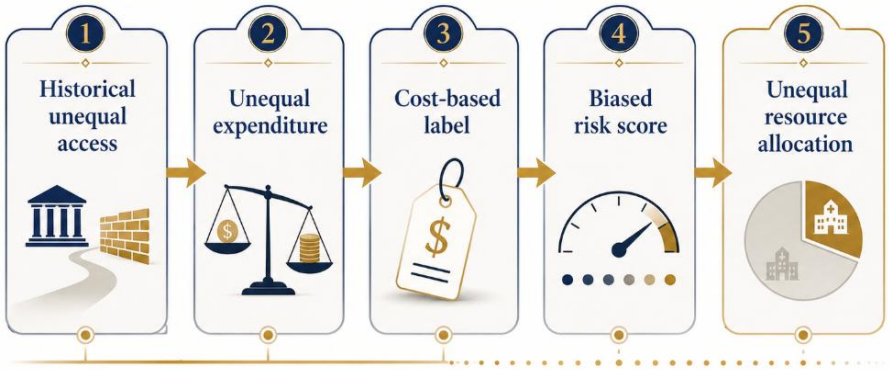



**Figure 21: Pathway from historical inequity to unequal resource allocation.**

If the model's recommendations affect future access, the system can create a feedback loop:

**Lower allocated care → Lower recorded spending → Lower future predicted need**

### Data-governance failures

- **Purpose mismatch:** The stated decision concerned health need, but the label measured expenditure.
- **Proxy not documented:** Users may interpret the score as clinical risk even though it predicts cost.
- **Insufficient subgroup audit:** Overall predictive performance did not reveal unequal meaning of the target across groups.
- **Administrative data treated as neutral:** Billing records reflected access, pricing, insurance, and treatment patterns.
- **Inadequate stakeholder analysis:** Patients affected by reduced resource allocation had limited visibility into the system.

### Redesign

- **Replace the target**
  Use direct or composite measures of health need, such as:
  - number and severity of chronic conditions;
  - physiological indicators;
  - preventable hospitalization;
  - uncontrolled disease;
  - functional status;
  - clinician-reviewed need.

- **Evaluate proxy validity**
  - For each subgroup $a$, examine $\mathbb{E}[H|C = c, A = a]$.
  - If the same cost corresponds to different need levels, cost is not a group-invariant proxy.

- **Perform subgroup calibration**
  - A score should have comparable clinical meaning:
    $$P(H \geq h_0|S = s, A = a)$$
    across relevant groups.

- Among people receiving the same model score $S = s$, what proportion **actually have a health need** above a **clinically important threshold** $h_0$, within each group.

- **Include governance review**
  The target variable should be approved jointly by:
  - clinicians;
  - data scientists;
  - ethicists;
  - affected communities;
  - operational leaders.

- **Monitor allocation outcomes**
  Measure:
  - who receives support;
  - unmet need;
  - health outcomes;
  - score distributions;
  - appeal or review rates;
  - subgroup differences.

### Why "remove race" is insufficient

Even if race is excluded, the model learns from variables shaped by structural inequality:

- prior utilization;
- insurance;
- location;
- diagnosis history;
- expenditure;
- access patterns.

Fairness requires examining how the target and features were generated, not merely deleting one sensitive column.

### Central lessons

- Accurate prediction of the wrong target is failure.
- Proxies must be validated against the real construct.
- Administrative records contain institutional inequalities.

- Fairness depends on target definition as well as model design.
- Sensitive attributes may be required to discover bias.
- Allocation systems require outcome and subgroup monitoring.
- Data governance must include conceptual review, not only security.

This case shows why **ethical data work is inevitable**: the **decisive error occurred before** model fitting, when healthcare cost was chosen to represent healthcare need.

### 2.11.3 Label Errors and Benchmark Instability

#### Background

Machine-learning research often selects models using performance on public benchmark test sets. This assumes that test labels are sufficiently correct to establish which model is superior.

Northcutt, Athalye, and Mueller investigated ten widely used datasets spanning computer vision, natural-language processing, and audio. They identified widespread test-label errors and estimated at least 3.3% errors on average across the ten datasets, including at least 6% in the ImageNet validation set. They showed that benchmark conclusions can change when evaluation labels are corrected or when mislabeled cases become more prevalent.

#### Why benchmark labels matter

Suppose true labels are $Y$, but the test set contains observed labels $\tilde{Y}$. Reported accuracy is:

$$\widehat{\mathrm{Acc}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{Y}_i = \tilde{Y}_i\}$$

The desired quantity is:

$$\mathrm{Acc}^* = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{Y}_i = Y_i\}$$

When $\tilde{Y}_i \neq Y_i$ a correct prediction may be counted as wrong, and an incorrect prediction may be counted as correct.

#### Sources of label error

- ambiguous classes;
- low-resolution images;
- multiple valid objects;
- annotator mistakes;
- ontology mismatch;
- outdated category definitions;
- automated labeling;
- transcription errors;
- context-dependent interpretation.

Some cases may not have one unambiguously correct label. This should be documented as uncertainty rather than hidden behind a forced category.

### Why better models can appear worse

- High-capacity models may learn fine-grained visual or semantic distinctions and disagree with erroneous test labels.
- A simpler model may reproduce common annotation patterns and score better against the noisy benchmark.
- Thus: agreement with benchmark labels ≠ agreement with reality.
- As benchmark scores become close, a small number of incorrect labels can alter rankings.

### Data-management failure

Removing any of these components **weakens reliability**.

- **Test sets treated as immutable truth:** Prestigious benchmark status discouraged continuous quality review.
- **Error reporting lacked a formal pathway:** Users may discover questionable labels without a standardized correction process.
- **Versioning problems:** Correcting labels without version control could make historical results incomparable.
- **Ambiguity not represented:** Single labels concealed uncertainty or multiple acceptable interpretations.

### A robust benchmark-governance process

Removing any of these components **weakens reliability**.

- **Multi-stage annotation**

    - initial annotation;
    - independent verification;
    - expert review for difficult cases;
    - adjudication;
    - uncertainty flagging.

- **Label confidence**
  Store $p_{ik} = P(Y_i = k|\text{annotations})$ rather than only a hard class where appropriate.

- **Challenge sets**
  Create subsets for:
    - rare classes;
    - ambiguous cases;
    - distribution shifts;
    - low-quality inputs;
    - near-duplicate classes.

- **Error-reporting mechanism**
  Dataset users should be able to submit:
    - record identifier;
    - suspected issue;
    - evidence;
    - proposed correction.

- **Versioned releases**
  Use semantic versions: $D^{(1.0)}, D^{(1.1)}, D^{(2,0)}$. Every paper should report the exact dataset version.

- **Statistical uncertainty**
  Report confidence intervals and test whether apparent performance differences exceed label uncertainty.

### Leakage and benchmark reuse

Public test sets face another risk: **repeated human** and **automated adaptation**. If researchers repeatedly inspect test errors and redesign models, the **community gradually tunes** to the test set **even without directly** training on its labels. The test set becomes part of the development process:

Test results → Research decisions → New models → Same test set

Possible responses include:

- hidden test sets;
- rotating evaluations;
- fresh data collection;
- external validation;
- limited submission frequency;
- post-deployment evaluation.

### Central lessons

- Test sets are datasets and require the same quality controls as training sets.
- Benchmark labels are measurements, not unquestionable truth.
- Small label-error rates can matter when model scores are close.
- Dataset versioning is necessary for scientific reproducibility.
- Annotation uncertainty should be represented where possible.
- Model selection should not depend on one noisy benchmark.
- Continuous curation is an essential research infrastructure function.

This case demonstrates the **inevitability of maintenance**: a dataset is not finished when first released. It requires correction, versioning, documentation, and governance throughout its useful life.

# Reading List

To support students in furthering their understanding of the topics covered in this module, here's a curated reading list organized by topic. This list includes books and research papers that delve deeper into each of the key areas discussed in this module.

### Data as a Constructed Record of Reality


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### Data Sources: Surveys, Sensors, Web Data, Logs, Administrative Records, and Experiments


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### Duplicate Records, Entity Resolution, and Record Linkage


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# Practice Questions for Quizzes and/or Exams

Please find below a list of **practice questions** to help you review **key concepts** and prepare for quizzes and/or exams.

### Data as a Constructed Record of Reality

1. Why is the phrase raw data considered misleading in data science and artificial intelligence?
2. Explain why a dataset should be understood as an engineered artifact rather than a passive reflection of reality.
3. How do conceptualization, observation, sampling, institutional context, and data representation jointly produce a dataset?
4. Why are labels such as disease, fraud, dropout, risk, and success not self-defining?
5. How can two institutions use the same variable name while assigning fundamentally different meanings to it?
6. Explain how operational definitions influence the target labels used in supervised learning.
7. What does the formulation $Q(D, T, C)$ communicate about dataset quality?
8. Why can a dataset be suitable for descriptive reporting but unsuitable for causal inference?
9. How can an exclusion decision during sampling affect the population to which a model can validly be applied?
10. Explain why the correct question is not Is this data good, but Is this data fit for the intended task and context?

### Data Sources and Their Inherent Realities

11. What are the principal advantages of surveys as a source of data for AI systems?
12. Distinguish among coverage error, sampling error, nonresponse bias, recall bias, social-desirability bias, and instrument effects.
13. How can a model trained on survey data learn reporting behavior rather than the intended real-world construct?
14. Explain the roles of systematic bias, random noise, environmental conditions, and device characteristics in sensor measurements.
15. What data-quality problems can result from calibration drift, clock misalignment, packet loss, firmware changes, and signal saturation?
16. Why should a social media platform be treated as part of the measurement instrument when web data are collected?

17. How can recommendation algorithms, moderation policies, bots, and duplicate posts distort conclusions drawn from web data?
18. Distinguish among event time, recording time, arrival time, and processing time in system logs.
19. Why do administrative records often represent institutional behavior and access patterns rather than objective ground truth?
20. Compare the strengths and limitations of controlled experiments, public datasets, and synthetic datasets?

## Public and Synthetic Data

21. Why does public availability not guarantee that a dataset is accurate, representative, ethically usable, or free of legal restrictions?
22. What is benchmark saturation, and how can repeated use of a public test set weaken the credibility of model comparisons?
23. How can label errors in a benchmark test set alter the ranking of competing machine learning models?
24. Why should public datasets undergo continuous auditing, correction, versioning, and maintenance?
25. What are the main methods used to generate synthetic data?
26. Identify six useful applications of synthetic data in machine learning and data engineering.
27. Why does synthetic data not automatically guarantee privacy or anonymity?
28. Define fidelity, utility, privacy, fairness, coverage, and realism as validation dimensions for synthetic data.
29. How can synthetic data reproduce or amplify biases contained in the original source data?
30. What evidence would be required before a model trained on synthetic clinical data could be trusted on real patients?

## Data Quality Dimensions

31. Define cell-level completeness and record-level completeness.
32. A variable contains 80 missing values among 1,000 records. Compute its cell-level completeness.
33. Why can a dataset with 99 percent populated cells still be incomplete for a particular application?
34. How should accuracy be assessed for numerical measurements and categorical labels?

35. Why does agreement among values within a database not establish that the values are correct?
36. Distinguish among formatting consistency, logical consistency, structural consistency, and cross-system consistency.
37. Explain the differences among freshness, latency, update frequency, and validity period.
38. What is feature staleness, and how can it reduce the reliability of real-time predictions?
39. Explain the difference between covariate shift, label distribution shift, and concept shift.
40. Why does a very large sample size not compensate for systematic underrepresentation of important populations?

### Measurement Error and Construct Validity

41. Distinguish between random measurement error and systematic measurement error.
42. Why can averaging reduce the effect of random error but fail to remove systematic error?
43. What is differential measurement error, and why is it important for fairness analysis?
44. How can a medical sensor that performs differently across skin tones create unequal model performance?
45. Define construct validity in the context of data collection and AI development.
46. Why can healthcare spending be a poor proxy for health need?
47. Why can arrest records be a poor proxy for actual crime commission?
48. Why can user clicks be a poor proxy for genuine user satisfaction?
49. Explain why model optimization cannot correct a target variable that measures the wrong construct.
50. How should domain experts evaluate whether a proposed proxy faithfully represents the concept of interest?

### Data Profiling and Curation

51. Define data curation and explain how it differs from data cleaning.
52. What activities are included in a complete data-curation lifecycle?
53. Describe the eight stages of the data governance and quality workflow presented in the module.
54. What information should be recorded in a data inventory before any modification begins?
55. What is data profiling, and why should it be performed before cleaning?
56. Which statistical, structural, temporal, and subgroup characteristics should be examined during profiling?

57. Distinguish among schema validation, relational validation, logical validation, and domain validation.
58. Why must original source data be preserved in an immutable state?
59. What information should be recorded in an audit trail for every correction or transformation?
60. How do version control, release approval, archiving, and retirement support long-term dataset reliability?

### Missing Values, Outliers, Duplicates, and Labels

61. List common reasons why values may be missing from real-world datasets.
62. Define Missing Completely at Random, Missing at Random, and Missing Not at Random.
63. Why are missingness mechanisms assumptions about the data-generating process rather than directly observable properties?
64. Compare deletion, simple imputation, groupwise imputation, multiple imputation, model-based imputation, and missing-value indicators.
65. Why must imputation parameters be estimated only from the training partition?
66. How can mean imputation reduce variance and distort relationships among variables?
67. Why should an outlier be investigated rather than automatically removed?
68. Compare the z-score rule, the interquartile-range rule, Winsorization, transformations, and robust loss functions.
69. Distinguish exact duplicates, near duplicates, false merges, and false splits in entity resolution.
70. Why should annotation systems preserve original labels, consensus labels, adjudication decisions, uncertainty, and acceptable alternative labels?

### Unit Errors, Class Imbalance, and Probability Calibration

71. Why should every numerical variable include metadata about units, valid ranges, precision, and conversion rules?
72. How can mixing kilograms and pounds in one variable produce serious analytical errors?
73. What is dimensional consistency, and why is it important when combining physical quantities?
74. Define class prevalence and class imbalance.
75. Why can a classifier that always predicts the majority class achieve high accuracy while having no practical value?
76. Compare class weighting, threshold adjustment, oversampling, undersampling, minority-class collection, and cost-sensitive learning.

77. Why must oversampling be performed only within the training partition or within each training fold?
78. What leakage can occur when synthetic minority observations are generated before train–test partitioning?
79. Why can a model trained on an artificially balanced dataset produce poorly calibrated probabilities at deployment?
80. Compare Platt scaling and isotonic regression as approaches to probability calibration.

### Data Leakage and Evaluation Design

81. Define data leakage and explain why it creates overly optimistic estimates of model performance.
82. What practical question should be asked when evaluating whether a feature introduces leakage?
83. Define target leakage and provide an example involving sepsis prediction.
84. What is train–test contamination, and through which preprocessing or modeling activities can it occur?
85. Why must scaling, imputation, encoding, feature selection, dimensionality reduction, and resampling occur inside each cross-validation training fold?
86. Compare standard $k$-fold, stratified $k$-fold, grouped cross-validation, and time-series cross-validation.
87. What is temporal leakage, and why is information-availability time more important than event time?
88. What is group leakage, and why must all records from the same patient, customer, household, or device remain in one partition?
89. When should an entire institution or site be held out during evaluation?
90. What checks should be included in a comprehensive data-leakage prevention checklist?

### Documentation, Governance, Privacy, Fairness, and Ethics

91. What information should be included in a comprehensive Datasheet for Datasets?
92. How does a data dictionary differ from a dataset-level datasheet?
93. Distinguish between data provenance and data lineage.
94. What information should be recorded for every transformation in a data-lineage chain?
95. Why should consent documentation describe permitted purposes, linkage, commercial use, international transfer, withdrawal, and vulnerable populations?
96. How do consent, licensing, copyright, and privacy impose different constraints on dataset use?

97. Why should de-identification be treated as risk reduction rather than a guarantee of anonymity?
98. Explain how data minimization, access control, encryption, retention limits, audit logs, differential privacy, and federated learning can reduce privacy risks.
99. Why might sensitive attributes be excluded from prediction but retained under restricted access for fairness auditing?
100. How should data owners, stewards, ethics committees, institutional review bodies, technical teams, domain experts, and organizational leaders share responsibility for the lawful, fair, transparent, and accountable use of data?

Below is a list of **scenario-based questions** designed to help students apply the concepts from this module to real-world AI systems.

### S1: University Student Risk Prediction

A university plans to develop an AI system that identifies students who may leave their degree programs. The model will use GPA, attendance, library usage, tuition payment history, and online learning activity.

The project team defines dropout as failure to register for the next semester. However, this definition also labels students who transfer to another university, take an approved leave, experience temporary financial difficulties, or delay registration because of administrative problems as dropouts.

Attendance is recorded through classroom scanners, but some instructors use manual attendance sheets and others do not record attendance. Missing attendance is stored as zero. The training data include only undergraduate students from the main campus, although the university intends to deploy the system for online students, graduate students, and students at satellite campuses.

The model achieves 87 percent accuracy. University administrators propose automatically restricting scholarships for students classified as high risk.

**Questions**

- How has the target variable dropout been constructed in this case?
- What problems arise from storing missing attendance as zero?
- Is the dataset representative of the intended deployment population? Explain.
- Evaluate the dataset using the fitness-for-purpose formulation $Q(D, T, C)$.
- What ethical problem arises from using the prediction to restrict scholarships?
- What documentation should be included in a Datasheet for Datasets before deployment?

### S2: Public Opinion Analysis from Social Media

A government agency wants to estimate public support for a new transportation policy. It collects four million public social media posts containing selected keywords and trains a sentiment classifier to label each post as supportive, neutral, or opposed.

The dataset contains many reposts of a small number of highly popular messages. Several automated accounts publish the same statements thousands of times. The platform recommendation algorithm promotes emotionally charged content, while moderate posts receive limited visibility. Most active users are young people living in urban areas, but the agency describes the findings as national public opinion.

Halfway through data collection, the platform changes its recommendation system and begins promoting short video content instead of text. The number of collected text posts declines sharply. The research team interprets this decline as reduced public interest in the policy.

**Questions**

- Why should the platform be treated as part of the measurement instrument?
- Identify the representativeness problems in the dataset.
- How can reposts and automated accounts distort the estimated level of public support?
- Does the decline in text posts necessarily represent a real change in public opinion?
- What legal and ethical issues should be examined before collecting and retaining the posts?
- What additional data source could provide a more defensible estimate of national opinion?

### S3: Hospital Readmission Model with Missing Data

A hospital develops an AI model to predict whether a patient will be readmitted within 30 days. The dataset includes age, diagnoses, previous admissions, blood pressure, laboratory results, length of stay, and discharge information.

Blood pressure is missing for 8 percent of patients. Lactate values are missing for 75 percent of patients because clinicians usually order the test only when severe illness is suspected. The data team replaces every missing numerical value with the overall mean calculated from the full dataset before creating the training and test partitions.

A profiling report also identifies several blood pressure values above 300 mmHg. The team deletes every extreme value automatically. It later discovers that some values were data-entry errors, while others came from patients in medical emergencies.

**Questions**

- Why might missing lactate values contain clinically meaningful information?
- Which missingness mechanisms could plausibly explain the absent blood pressure and lactate values?
- What statistical problems can result from mean imputation?
- How has the team introduced train–test contamination?
- Why should the extreme blood pressure observations be investigated rather than automatically deleted?
- Propose a more principled workflow for handling the missing values and outliers.

### S4: Cancer Detection with Imbalanced Medical Images

A research team trains an image classifier using scans from 1,000 patients. The dataset contains 950 patients without cancer and 50 patients with cancer. To balance the classes, the team applies SMOTE and image augmentation to the complete dataset before dividing it into training and testing sets.

Each patient has several images from the same examination. The images are randomly divided at the image level, so images from one patient may appear in both training and testing sets. The team also standardizes all images using statistics calculated from the full dataset.

The resulting model achieves 99 percent accuracy. However, when tested at a hospital that did not contribute training data, performance declines substantially. The model also produces very high cancer probabilities even though cancer prevalence in the deployment population is only 5 percent.

**Questions**

- Why is accuracy alone inadequate for evaluating this model?
- Identify three forms of data leakage or contamination in the development process.
- Why must all images from the same patient remain in one data partition?
- When should SMOTE or augmentation be applied?
- Why might the predicted probabilities require recalibration?
- What evaluation design would provide evidence of generalization to a new hospital?

### S5: Commercial Use of a Public Medical Dataset

A university research group develops an AI system for early detection of a neurological disease using a publicly accessible medical imaging dataset. The dataset page states that access is available for academic research. The original consent form permits analysis of medical images but prohibits commercial use and international transfer.

A technology company offers to purchase the model and deploy it in several countries. The research team argues that the images have been de-identified and that only the trained model, rather than the original images, will be transferred.

The researchers also combine the dataset with another openly licensed dataset and plan to publish the combined collection. Their documentation contains only the statements Consent obtained: Yes and Data publicly available. It does not record dataset versions, preprocessing steps, excluded cases, consent conditions, license restrictions, or lineage.

**Questions**

- Why is Consent obtained: Yes inadequate documentation?
- Distinguish among consent, licensing, copyright, and privacy in this case.
- Does de-identification automatically permit commercial or international use?
- What license-compatibility problem may arise when the two datasets are combined?
- What information should be added to the dataset documentation and data-lineage record?
- Which organizational roles should approve or reject the proposed commercial deployment?

These are **take-home reflection questions** intended to encourage **deep thinking and planning**. There are no single correct answers. Each question can't be answered in a couple sentences. It will push you to think deeply about the concepts, principles, and practices in this module.

### R1: A National Student Success and Intervention System

The national higher education authority plans to build an AI system that identifies university students at risk of leaving their programs. The purpose is to provide early academic, financial, and psychological support.

The participating universities submit the following data:

- GPA and course grades
- attendance records
- tuition payment history
- library and online learning activity
- disciplinary records
- scholarship status
- accommodation location
- counseling service usage
- family income
- disability status

However, the universities define dropout differently. Some label a student as a dropout after one missed semester. Others wait for two years. Some include students who transfer to another institution, while others do not. Attendance is recorded through scanners at certain universities, manual registers at others, and not recorded at all in some departments.

The training dataset contains mostly students from large urban universities. Students from rural campuses, online programs, small private institutions, and vocational colleges are poorly represented. The project team proposes replacing all missing values with the overall mean and removing students with unusual academic histories.

The final model assigns each student a risk score. Some universities intend to use the score to offer additional support. Others intend to reduce scholarships for students classified as highly likely to leave.

**Questions**

- Develop a precise operational definition of student dropout. Explain how your definition treats temporary leave, delayed registration, transfer, expulsion, late graduation, and administrative registration errors.
- Apply the fitness-for-purpose formulation $Q(D, T, C)$ to determine whether the available dataset is suitable for national deployment. Define the dataset, task, and context explicitly.
- Explain how the dataset represents a constructed institutional record rather than an objective description of student motivation or ability. Identify the human, technical, administrative, and policy decisions embedded in the data.
- Design a representativeness audit covering institution type, geographic region, mode of study, socioeconomic status, disability, gender, program level, and intersectional groups. Explain what evidence would justify national deployment.
- Propose a missing-data strategy for attendance, income, counseling usage, and online activity. For each variable, discuss whether missingness may be MCAR, MAR, or MNAR and whether the absence itself could be informative.
- Evaluate whether counseling usage and disability status should be included as predictive features, retained only for fairness auditing, or excluded entirely. Consider privacy, data minimization, potential benefit, and possible harm.
- Design a governance structure for the system. Specify the responsibilities of data owners, data stewards, university administrators, ethics committees, technical teams, student representatives, and national regulators.
- Compare the ethical consequences of using the risk score to provide assistance with using it to restrict scholarships. Explain how the same model can produce beneficial or harmful outcomes depending on the decision policy surrounding it.
- Prepare a plan for contestability and human review. Explain how students could inspect, challenge, and correct the data and decisions associated with their risk scores.

- Identify the conditions under which the project should be delayed, redesigned, limited to selected institutions, or rejected entirely.

### R2: A Smart City Predictive Policing Platform

A city government plans to predict where serious crime is likely to occur during the next week. The proposed system combines:

- historical arrest records
- emergency telephone calls
- police patrol logs
- reported crimes
- CCTV detections
- social media posts
- geographic and demographic information
- records from courts and correctional institutions

The developers treat arrest counts as a measure of crime prevalence. Neighborhoods with more recorded arrests receive higher predicted risk scores and consequently receive more patrol officers. Increased policing produces more stops, searches, and arrests, which are then added to the training dataset.

Residents argue that certain neighborhoods have historically received more police attention and therefore generate more records, even when underlying offending rates are similar. They also claim that the system may treat poverty, housing density, and ethnicity as indirect proxies for criminality.

The police department proposes retaining detailed location and identity information indefinitely because it may be useful for future investigations.

**Questions**

- Evaluate the construct validity of using arrests as a proxy for crime commission. Propose alternative outcome definitions and discuss the limitations of each.
- Map the complete data-generating process from underlying events to reported incidents, police responses, arrests, administrative records, and model predictions. Identify where institutional behavior shapes the observed data.
- Explain how a feedback loop could arise between historical policing, recorded arrests, predicted risk, and future police deployment. Discuss how this loop may reinforce historical inequalities.
- Determine which variables may act as proxies for sensitive attributes even if race or ethnicity is removed. Explain why removing explicit sensitive attributes does not necessarily eliminate discrimination.

- Design a fairness evaluation that considers geographic, demographic, socioeconomic, and intersectional disparities. Specify what outcomes should be compared and how unequal measurement rates should be handled.
- Propose a data-minimization policy for identity, location, social media, and CCTV data. Explain which information is necessary, which can be aggregated, and how long different data elements should be retained.
- Develop an oversight and accountability framework involving police leadership, courts, elected officials, community representatives, civil-rights experts, data stewards, and independent auditors.
- Discuss whether individuals should be informed that their neighborhoods or activities contributed to model training. Consider transparency, public safety, privacy, and procedural justice.
- Propose safeguards that prevent a model-generated risk score from becoming automatic justification for stops, searches, surveillance, or arrests.
- Decide whether predictive policing can be made ethically acceptable through technical and governance controls, or whether its underlying measurement and power structures create limitations that cannot be solved through model improvement alone.

### R3: A Multihospital Sepsis Early-Warning System

A healthcare network wants to build an AI system that predicts sepsis six hours before clinical recognition. Data are collected from twelve hospitals and include:

- vital signs
- laboratory results
- medication orders
- diagnosis codes
- clinical notes
- device measurements
- intensive care transfers
- mortality and discharge records

The system achieves an AUC of 0.98 during internal evaluation. Investigation shows that broad-spectrum antibiotic orders and intensive care transfer requests are among the strongest predictors. These variables often become available only after clinicians suspect sepsis.

Some hospitals use different laboratory equipment, measurement units, and electronic record systems. A device firmware update changes the distribution of heart-rate readings halfway through the collection period. Several patients appear in the dataset multiple times, and images or records from the same patient may enter different partitions.

Lactate values are missing for most patients because clinicians order the test mainly for those who appear seriously ill. The team proposes imputing all missing lactate values with the population mean. The training data contain few patients from rural hospitals and limited representation of some age and ethnic groups.

**Questions**

- Define the prediction time and outcome window precisely. Construct a timeline showing which variables are legitimately available six hours before clinical recognition.
- Identify possible target leakage, temporal leakage, group leakage, site leakage, and preprocessing contamination. Explain how each could inflate reported performance.
- Evaluate whether antibiotic orders and intensive care transfer requests should be included as predictors. Distinguish early physiological risk factors from variables that encode clinician recognition or response.
- Develop an appropriate partitioning and validation strategy. Consider patient-level grouping, chronological evaluation, hospital-level holdout, cross-validation, and a final untouched test set.
- Analyze the missing lactate values. Explain why missingness may be informative and how mean imputation could erase clinically meaningful information.
- Design a harmonization plan for different laboratory units, devices, firmware versions, sampling frequencies, clocks, and electronic health record systems.
- Propose a subgroup and intersectional evaluation plan. Include performance, calibration, false-negative rates, false-positive rates, and measurement quality across hospitals and populations.
- Create a data-lineage plan that connects every trained model to the exact source data, dataset versions, preprocessing code, parameters, validation results, and responsible personnel.
- Design a deployment policy explaining how the model output should interact with clinical judgment. Include alert thresholds, escalation procedures, override mechanisms, audit logs, and monitoring for alert fatigue.
- State the evidence that would be required before the model could be deployed at a hospital that did not contribute data to development.

### R4: Synthetic Financial Data for Credit Approval

A financial technology company wants to develop a credit approval model but faces restrictions on sharing customer records with external developers. It trains a generative model on historical loan applications and produces a synthetic dataset containing:

- income
- occupation
- age
- geographic region
- repayment history

- property ownership
- previous defaults
- loan amount
- approval decisions

The synthetic data closely match the overall distributions of the source dataset. External developers train a classifier that performs well when evaluated on a synthetic test set.

However, the original data contain few applicants from rural regions, informal employment sectors, and low-income households. Historical approval decisions may also reflect discriminatory lending practices. Rare default patterns are poorly represented. Several synthetic records are unusually similar to real customers, raising possible membership inference and reconstruction risks.

Management argues that because the dataset is synthetic, consent, privacy, fairness, and licensing concerns no longer apply.

**Questions**

- Challenge or defend the claim that synthetic data are automatically anonymous, fair, representative, and ethically unrestricted.
- Design a validation framework covering fidelity, utility, privacy, fairness, coverage, and realism. For each dimension, specify suitable tests and acceptable evidence.
- Explain why matching overall distributions is insufficient. Identify subgroup, intersectional, tail, and dependency structures that should also be examined.
- Analyze how historical discrimination in approval decisions could be reproduced or amplified by the synthetic generator and the downstream credit model.
- Propose methods for evaluating membership inference, reconstruction, memorization, and record similarity risks.
- Determine whether protected attributes should be included in the synthetic dataset, excluded from model training, or preserved under restricted access for fairness auditing.
- Explain why evaluation only on synthetic test data cannot establish real-world utility. Propose a secure method for evaluating models on representative real data.
- Develop a documentation package for the synthetic dataset, including motivation, source data, generation method, model version, privacy safeguards, known limitations, intended uses, discouraged uses, and maintenance procedures.
- Consider a case where the synthetic dataset improves privacy but reduces representation of already marginalized applicants. Explain how the organization should balance privacy, fairness, and utility.
- Recommend whether the synthetic dataset should be approved for internal experimentation, external research, operational credit decisions, or none of these uses. Justify different approval levels.

### R5: Commercial Reuse of an International Medical Imaging Repository

A consortium of universities creates a medical imaging repository using scans collected from several countries. The original consent forms vary considerably:

- some permit academic research only
- some permit commercial research
- some prohibit international transfer
- some permit future unspecified health research
- some allow withdrawal at any time
- some do not mention AI development

The dataset is released under a research-only license. Images are de-identified by removing names and identification numbers, but retain exact examination dates, rare diagnoses, age, sex, and hospital location.

A technology company proposes combining the repository with two open datasets and training a commercial diagnostic model. The company plans to distribute the model globally but not distribute the original images. It argues that model weights are not personal data and that open access to some of the source datasets makes the combined project permissible.

During development, the team discovers inconsistent diagnosis labels, undocumented exclusions, duplicated scans, incompatible licenses, and missing records of the preprocessing applied to earlier dataset versions.

**Questions**

- Distinguish consent, privacy, licensing, copyright, institutional approval, and ethical acceptability. Explain why satisfying one of these does not automatically satisfy the others.
- Develop a consent matrix that maps each data source to permitted purposes, commercial use, international transfer, withdrawal rights, redistribution, and AI model development.
- Analyze whether transferring model weights rather than images resolves the ethical and legal concerns. Consider memorization, membership inference, model inversion, and restrictions on derived products.
- Evaluate the re-identification risk created by exact dates, rare diagnoses, age, sex, and hospital location. Propose a risk-reduction strategy that preserves useful information where possible.
- Design a license-compatibility review for combining the three datasets. Explain how the most restrictive terms may affect redistribution and commercial deployment.

- Develop a plan for handling participant withdrawal after a model has already been trained. Consider deletion of source records, retraining, model updates, documentation, and communication with downstream users.
- Propose a quality-control workflow addressing duplicate scans, inconsistent labels, undocumented exclusions, missing provenance, and historical preprocessing.
- Prepare a minimum Datasheet for Datasets and data dictionary for the combined repository. Identify the information required before any external user could assess fitness for purpose.
- Design an approval process involving data custodians, ethics committees, legal experts, privacy officers, clinical specialists, technical teams, participant representatives, and organizational leadership.
- Make a final recommendation on whether the commercial project should proceed, proceed under restrictions, require renewed consent, use only selected data sources, or be rejected. State the assumptions, evidence, and unresolved uncertainties underlying your decision.